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Forecasting House Prices: The Role of Market Interconnectedness

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ABSTRACT

While the existing research uncovers interconnections between various housing markets, it largely ignores the question of whether such linkages can improve house price predictions. To address this issue, we proceed in two steps. First, we forecast disaggregated house price growth rates from Australia and China to determine whether incorporating price links can improve out-of-sample predictions. We find that accounting for within-city house price interconnectivity in Sydney and Melbourne can indeed improve house price predictions. However, when forecasting city-level prices from China, univariate models produce superior predictions. Second, in order to shed light on our empirical findings, we conduct simulation experiments calibrated to reflect the connections estimated from the data. The predictive ability of house price connectivity hinges on the sparsity and strength of the connections between interconnected markets. In the presence of stronger and denser connections, connectivity information is crucial for improving short-term forecasts. On the other hand, when the connections are sparse and weak (as in the Chinese housing data), the univariate models outperform. Our study shows that finding evidence of significant price interconnections does not always lead to forecasting gains.

JEL Classification: R39, C32, C51, C53

1 | Introduction

The residential real estate sector is a key driver of economic growth and affects many important macroeconomic variables. The literature shows that house prices influence consumption (Case et al. 2005; Campbell and Cocco 2007), employment (Abraham and Hendershott 1996), income (Case and Shiller 1990; Malpezzi 1999; Favilukis et al. 2017), interest rates (Muellbauer and Murphy 1997), and inflation (Abelson et al. 2005). Furthermore, financial markets transmit shocks from the housing sector to other seemingly unrelated parts of the economy, as evidenced by the worldwide financial meltdown of 2007. Given the important role that house prices play in a globally connected economy, they are closely monitored, and accurate predictions of future real estate market conditions are sought after by both private and public sector institutions. In this paper, we extend the literature on forecasting house

prices by examining whether utilizing the interconnectedness between disaggregated house prices can improve the accuracy of their out-of-sample forecasts.

An important empirical finding is that house prices exhibit connectivity across various domains. The observation that house prices move in tandem has been tested on national (Hirata et al. 2013; Milunovich 2020b), regional (DeFusco et al. 2018; Gupta and Miller 2012; Flor and Klarl 2017) and within-city (Hurn et al. 2022) levels. Such relationships can form through a number of different mechanisms, including macroeconomic and financial market dependencies, geographic and cultural proximity, as well as through regional development plans in the case of within-metropolitan connectivity (Carleton et al. 2022). Although existing research provides ample evidence for the presence of interconnectedness in various housing markets, see also Cotter et al. (2015), Tsai (2015), Zhang and Fan (2019) and

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Antonakakis et al. (2018), the literature largely ignores the question of whether uncovered connections can improve house price forecasts.

The literature on modeling and predicting house prices is extensive and has been reviewed in several surveys, including Gatzlaff and Tirtiroğlu (1995), Ghysels et al. (2013), Duca et al. (2021), and Meszaros (2024). Although a comprehensive review of this work is beyond the scope of our paper, we highlight several findings that are particularly relevant to our discussion. The focus of early research on the predictability of house prices was on testing market efficiency and assessing the effectiveness of various predictors based on in-sample analysis. The key finding of this literature is that real house price growth rates exhibit positive serial correlation and predictability over short horizons (e.g., Case and Shiller 1989, 1990; Gau 1984, 1985; Schindler 2013). This predictability is linked to various measures of income, interest rates, population, inflation, rent, and labor market variables (see, e.g., Abraham and Hendershott 1996; Cho 1996; Johnes and Hyclak 1999). However, as noted in Rapach and Strauss (2007, 2009), it is difficult to choose a priori a set of variables that are consistently effective for out-of-sample house price forecasting because their predictive power can vary over time. Next, there is evidence to suggest that large-scale models tend to outperform smaller specifications in forecasting house price growth (e.g., Gupta 2013), especially models accounting for geographic proximity (e.g., Gupta and Das 2010), economic fundamentals (e.g., Bork and Müller 2018), and financial liquidity factors (e.g., Ellington et al. 2023).

In this paper, we analyze within-city house prices in Australia and city-level prices in China, directly addressing the question of whether accounting for house price connections can help improve their out-of-sample predictions. We make a twofold contribution to the literature. First, in our empirical analysis, we study house price growth rates for Local Government Areas (LGAs)¹ within Sydney (32 LGAs) and Melbourne (31 LGAs), the two largest cities in Australia, and for city-level house price growth rates across 88 cities in China. House price within these markets is analyzed in Hurn et al. (2022), who find evidence of increasing price connectedness over time. However, they do not explore whether this connectedness can enhance out-of-sample predictions of disaggregated house prices, compared to models that do not consider connectivity. To answer this question, we capture within-region price connectivity using large-dimensional vector autoregressive (LDVAR) models separately for each of the three markets. Due to the high number of parameters present in these models, we employ the least absolute shrinkage and selection operator (LASSO) technique for coefficient estimation.

The combined LDVAR-LASSO method produces sparse models, where only a subset of the coefficients are nonzero and thus functions as a variable selection device. Our LDVAR-LASSO approach is related to the NETS framework of Barigozzi and Brownlees (2019). However, while NETS imposes sparsity on both the VAR dynamics and the inverse covariance matrix of the innovations to uncover contemporaneous network structures, our approach imposes sparsity only on the VAR coefficients. Such models have been shown to produce good forecasting performance in macroeconomic applications; see, for example, De Mol et al. (2008), and in finance, see, for example, Chinco

et al. (2019). While the LDVAR-LASSO method has been employed to evaluate connectedness among house price growth series in Gabauer et al. (2024) and Balcilar et al. (2022), it has not yet been used to determine whether the observed connectedness can enhance out-of-sample predictions in this context. Thus, we specify the models with and without connectivity and measure the impact of accounting for price interdependencies on the prediction accuracy. We generate out-of-sample forecasts for 1-, 3-, 6-, and 12-months ahead house price growth rates and employ the squared forecast error loss function and the model confidence set (MCS) procedures of Hansen et al. (2011) to evaluate the predictions.

Our second contribution to the literature expands the understanding of LDVAR-LASSO methods in the context of out-of-sample forecasting and is thus largely methodological in nature. Namely, we are interested in uncovering the conditions under which house price interconnectedness can improve out-of-sample forecasts. Since we capture connectedness via LDVAR models, we propose investigating the effects of sparsity and connectivity strength of the LDVAR coefficient matrices on predictive efficacy. In particular, we define sparsity by the proportion of zero elements in the parameter matrices, while connectivity strength is gauged by the magnitude of the lagged connections. Although the statistical properties of LASSO estimators for LDVAR models have been examined in the literature, see, for example, Basu and Michailidis (2015) and Kock and Callot (2015), there are few systematic investigations of their out-of-sample forecasting performance and the relationship with the underlying interdependency structure. We address this issue by conducting a simulation study calibrated to our empirical results.

We summarize our results as follows. The LDVAR model outperforms univariate models across all horizons for Sydney and at short horizons (1–6 months) for Melbourne, while univariate models consistently dominate in the Chinese housing markets. Robustness checks with alternative model specifications and evaluation criteria confirm that our results remain largely unchanged. Our simulation study highlights several key findings. While both sparsity and connectivity strength affect the predictive performance of LDVAR models, connectivity strength plays the dominant role. When connectivity is weak, univariate models outperform LDVAR across all horizons. When connectivity is strong, ignoring interdependence leads to poorer forecasts, though the impact diminishes at longer horizons. These results are robust to different loss functions (squared and absolute errors) and estimation methods (LASSO and adaptive LASSO).

The paper is structured as follows. Section 2 outlines our modeling approach, while Section 3 introduces the data and presents our primary empirical findings. This discussion is expanded in Section 4, where we consider a number of alternative specifications and robustness tests. In Section 5, we provide an explanation for our empirical findings by conducting a simulation study. Finally, Section 6 contains our concluding remarks.

2 | The Econometric Method

This section presents our empirical framework for incorporating connectivity into large-dimensional multivariate time series

forecasting and evaluating its effectiveness in improving out-of-sample predictions. In Section 2.1, we introduce the dynamic forecasting models utilized in our study, and in Section 2.2, we elaborate on the estimation methods used to fit these models. After estimating the models, we generate out-of-sample predictions and assess their performance using the techniques detailed in Section 2.3. Following our main analysis using these methods, we perform several robustness checks, as explained in Section 2.4. Lastly, we conduct a simulation study, calibrated to our empirical findings, to explore the impact of different levels of connectedness on the predictive effectiveness of our models, as outlined in Section 2.5.

2.1 | Forecasting Models

Let $\mathbf{y}_t = [y_t^{(1)}, y_t^{(2)}, \dots, y_t^{(N)}]'$ be a N -dimensional multivariate time series of the variable of interest in period t , $t = 1, \dots, T$ and $y_t^{(i)}$, $i = 1, \dots, N$, denote the i th element of $\mathbf{y}_t$. In our empirical applications, $y_t^{(i)}$ represents the real housing growth rate/return at period t for the cross-section unit (i.e., city or LGA) i . It is computed from the prices $P_t^{(i)}$ as follows $y_t^{(i)} = 100 \times \ln(P_t^{(i)}/P_{t-1}^{(i)})$.

We use two univariate models as benchmarks to judge the performance of more complicated models against. The first one is the autoregressive (AR) model, defined as

$$y_t^{(i)} = \phi_0^{(i)} + \sum_{j=1}^p \phi_j^{(i)} y_{t-j}^{(i)} + \varepsilon_t^{(i)}, \quad (1)$$

while the second model is the autoregressive moving average (ARMA) model, given by

$$y_t^{(i)} = \phi_0^{(i)} + \sum_{j=1}^p \phi_j^{(i)} y_{t-j}^{(i)} + \sum_{k=1}^q \theta_k^{(i)} \varepsilon_{t-k}^{(i)} + \varepsilon_t^{(i)}, \quad (2)$$

where $\phi_j^{(i)}$ are the parameters of AR terms and $\theta_k^{(i)}$ are the parameters of moving average (MA) terms. In particular, the AR(1) and ARMA(1,1) models have proven to be robust choices for predicting housing returns (Crawford and Fratanoni 2003; Rapach and Strauss 2009; Balcilar et al. 2015). It is important to highlight that the forecasting models mentioned above rely solely on the i th element of $\mathbf{y}_t$ for predictions and do not take into account the influence of cross-market links represented by the other elements in $\mathbf{y}_t$.

The vector autoregressive (VAR) model is an extension of the univariate AR model and is commonly employed as a multivariate time series model in macroeconomics and finance. In contrast to the AR(p) model, the VAR(p) model allows for the incorporation of temporal interdependencies among the constituent time series in $\mathbf{y}_t$. It is defined as

$$\mathbf{y}_t = \Phi_0 + \sum_{j=1}^p \Phi_j \mathbf{y}_{t-j} + \varepsilon_t \quad (3)$$

$$= \Phi_0 + \Phi \mathbf{Z}_t + \varepsilon_t, \quad (4)$$

where $\Phi_0 = [\phi_0^{(1)}, \phi_0^{(2)}, \dots, \phi_0^{(N)}]'$ is an N -dimensional intercept vector allowing for nonzero means, ε_t represents an N -dimensional residual vector with nonsingular covariance matrix Σ_ε , and $\mathbf{Z}_t = [\mathbf{y}_{t-1}', \mathbf{y}_{t-2}', \dots, \mathbf{y}_{t-p}']'$ is a Np -dimensional vector that contains all the lagged $\mathbf{y}_t$. Φ_j is a $N \times N$ endogenous parameter matrix at lag j , $j = 1, 2, \dots, p$, defined as

$$\Phi_j = \begin{bmatrix} \Phi_j^{(1)} \\ \Phi_j^{(2)} \\ \vdots \\ \Phi_j^{(N)} \end{bmatrix} = \begin{bmatrix} \phi_{1j}^{(1)} & \phi_{2j}^{(1)} & \dots & \phi_{Nj}^{(1)} \\ \phi_{1j}^{(2)} & \phi_{2j}^{(2)} & & \phi_{Nj}^{(2)} \\ \vdots & & \ddots & \vdots \\ \phi_{1j}^{(N)} & \phi_{2j}^{(N)} & \dots & \phi_{Nj}^{(N)} \end{bmatrix}, \quad (5)$$

and $\Phi = [\Phi_1, \Phi_2, \dots, \Phi_p]$ is a $N \times Np$ parameter matrix. While the off-diagonal elements of Φ_j are crucial for comprehending the intertemporal connections within the data, the high level of parameterization required can potentially hinder the practical usefulness of the VAR model.

To isolate the effect of connectivity between the various components of $\mathbf{y}_t$, we break down the matrix Φ_j into two parts: Φ_j^d , which contains the diagonal elements of Φ_j representing the influence of each variable's own lagged information, and Φ_j^{od} , which includes the off-diagonal elements that capture the interactions between different elements in $\mathbf{y}_t$. These matrices are defined as follows:

$$\Phi_j^d = \begin{bmatrix} \phi_{1j}^{(1)} & 0 & \dots & 0 \\ 0 & \phi_{2j}^{(2)} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & \phi_{Nj}^{(N)} \end{bmatrix} \text{ and } \Phi_j^{od} = \begin{bmatrix} 0 & \phi_{2j}^{(1)} & \dots & \phi_{Nj}^{(1)} \\ \phi_{1j}^{(2)} & 0 & & \phi_{Nj}^{(2)} \\ \vdots & & \ddots & \vdots \\ \phi_{1j}^{(N)} & \phi_{2j}^{(N)} & \dots & 0 \end{bmatrix}.$$

By separating the diagonal and off-diagonal elements of Φ_j as above, the VAR(p) model given in (3) can be decomposed as

$$\mathbf{y}_t = \Phi_0 + \underbrace{\sum_{j=1}^p \Phi_j^d \mathbf{y}_{t-j}}_{\text{Autoregressive Effects}} + \underbrace{\sum_{j=1}^p \Phi_j^{od} \mathbf{y}_{t-j}}_{\text{Connectivity Effects}} + \varepsilon_t. \quad (6)$$

This model specification shares similarities with the one explored in Zhu et al. (2017). In their study, Zhu et al. (2017) link the intercepts to unit-specific (time-invariant) exogenous variables and assume that the system affects unit i through an aggregated quantity defined as follows: $\bar{y}_{t-j}^{(i)} = \frac{1}{n_i} \sum_{m=1}^N w_{mj}^{(i)} y_{t-j}^{(m)}$, where $w_{sj}^{(i)}$ represents the connection strength between unit m and i at period $t-j$ and assumed to be known and $n_i = \sum_{s \neq i} w_{sj}^{(i)}$.² In contrast, we refrain from imposing those constraints on model coefficients to fully leverage the flexibility of VAR for forecasting. Furthermore, we extend our model to include exogenous variables capturing observable unit-specific and common factors in Section 4.1.

While VAR(p) model, as shown in equation (4), can be estimated through multivariate least squares in low-dimensional settings, that is, where the number of predictors is smaller than the time series length T (Lütkepohl 2005), estimation becomes challenging in large-dimensional settings. This leads to the so-called “curse of dimensionality” which in the context of LDVAR models relates to a loss of degrees of freedom and poor forecasting performance caused by the large number of parameters to be estimated (De Mol et al. 2008). As demonstrated in Equation (5), the parameter space of VAR grows quadratically with the dimension of $\mathbf{y}_t$ and linearly with the VAR order, leading to a large number of parameters even for moderate dimensions. In such cases, standard estimation techniques like least squares become computationally infeasible.

Various methods have been proposed in the literature to tackle the issue of high dimensionality in VAR models. These methods include reduced-rank techniques (Reinsel 1983; Ahn and Reinsel 1988; Carriero et al. 2011), factor models (Canova and Ciccarelli 2009; Stock and Watson 2016; Koop and Korobilis 2019), Bayesian methods (Karlsson 2013; Koop 2017; Koop and Korobilis 2016), and penalized estimation of sparse VAR (Wilms and Croux 2016; Davis et al. 2016; Smeekes and Wijler 2018; Camehl 2023). Among these methods, the penalized estimation approach has gained significant attention due to its ability to handle large-dimensional data and identify relevant predictors. In this study, we use the LASSO estimator introduced by Tibshirani (1996) to estimate the LDVAR model and uncover the underlying system structure. LASSO can handle a large number of predictors, produces sparse and interpretable models, and has been shown to have good forecasting performance in various applications. We have also explored the adaptive LASSO method introduced by Zou (2006). However, in our time series simulation settings, the LASSO estimator consistently outperformed the adaptive LASSO estimator. Therefore, we concentrate our discussion on the LASSO estimator and relegate the outcomes of utilizing the adaptive LASSO in the Appendix. The next subsection introduces the LASSO estimation method.

2.2 | LASSO Estimation

LASSO has been employed as a model selection device to maintain relevant variables while removing irrelevant predictors under various model specifications (Leeb and Pötscher 2005; Zhao and Yu 2006; Meinshausen and Bühlmann 2006; Bickel et al. 2009; Loh and Wainwright 2011; Masini et al. 2022; Masini et al. 2021). In particular, Nardi and Rinaldo (2011) showed its estimation, selection, and prediction consistency for AR models, while Loh and Wainwright (2011), Kock and Callot (2015) and Basu and Michailidis (2015), provided the nonasymptotic estimation and prediction error bounds and proved the estimation and prediction consistency for the LASSO in Gaussian VAR models.

It is conventional to apply LASSO estimation equation by equation within the VAR system, as opposed to applying it to the entire system in one step (Kock and Callot 2015; Medeiros and Mendes 2016; Han and Tsay 2020; Masini et al. 2022). This technique has the potential to enhance estimation accuracy while reducing the dimensionality of the parameter space. We adhere

to this established practice. To illustrate the mechanism of LASSO, we rewrite (4) for the unit i as follows:

$$\mathbf{y}^{(i)} = \boldsymbol{\iota} \phi_0^{(i)} + \mathbf{Z} \boldsymbol{\Phi}^{(i)} + \boldsymbol{\varepsilon}^{(i)}. \quad (7)$$

In this equation, $\mathbf{y}^{(i)} = [y_1^{(i)}, y_2^{(i)}, \dots, y_T^{(i)}]'$ contains all the observations for i th variable, $\boldsymbol{\iota}$ is a T -dimensional vector of one, $\boldsymbol{\Phi}^{(i)} = [\Phi_1^{(i)}, \dots, \Phi_p^{(i)}]'$ is a Np -dimensional coefficient vector, $\mathbf{Z} = [\mathbf{Z}_1, \dots, \mathbf{Z}_T]'$ is of dimensions $T \times Np$, and $\boldsymbol{\varepsilon}^{(i)} = [\varepsilon_1^{(i)}, \dots, \varepsilon_T^{(i)}]$ is a T -dimensional vector of error terms.

The number of parameters in the vector $\boldsymbol{\Phi}^{(i)}$ might significantly exceed the sample size, but only a subset of these parameters may actually be nonzero, resulting in a sparse true parameter vector denoted as $\boldsymbol{\Phi}^{*(i)}$. This means that the connectivity between the variable $\mathbf{y}^{(i)}$ and other variables $\mathbf{y}^{(j)}$, where $j \neq i$, may exist only between certain variables or at specific lags. In this scenario, applying LASSO becomes essential for selecting non-zero coefficients, effectively identifying the most relevant variables for each equation within the VAR model and enhancing forecasting accuracy. The LASSO estimator for Equation (7) is obtained by solving

$$[\hat{\phi}_0^{(i)}, \hat{\boldsymbol{\Phi}}^{(i)}]' = \underset{\phi_0^{(i)}, \boldsymbol{\Phi}^{(i)}}{\operatorname{argmin}} \frac{1}{2T} \|\mathbf{y}^{(i)} - \boldsymbol{\iota} \phi_0^{(i)} - \mathbf{Z} \boldsymbol{\Phi}^{(i)}\|_{\ell_2}^2 + \lambda \|\boldsymbol{\Phi}^{(i)}\|_{\ell_1}, \quad (8)$$

where $\|a\|_{\ell_1} = \sum_{i=1}^n |a_i|$ and $\|a\|_{\ell_2} = \sqrt{\sum_{i=1}^n a_i^2}$, respectively, denote the ℓ_1 and ℓ_2 norms, for any vector $a \in \mathbb{R}^n$. Equation (8) can be regarded as a least squares objective function augmented with an additional term that penalizes coefficients in $\boldsymbol{\Phi}^{(i)}$. Here, λ is a nonnegative tuning parameter, which can be estimated through cross-validation or information criteria. The intercept $\phi_0^{(i)}$ is not penalized and assumed to be nonspare.

Several papers have explored the nonasymptotic upper bounds for prediction and estimation errors of LASSO for the VAR model (3) under different model assumptions. Among them,

Kock and Callot (2015) assumes that all roots of $\left|I_N - \sum_{j=1}^p \boldsymbol{\Phi}_j z^j\right|$ lie outside the unit circle and the error term ε_t follows an independent and identically distributed Gaussian distribution. On the other hand, Basu and Michailidis (2015) presents a more general assumption that allows both the predictors and the errors to be generated from stable Gaussian processes. Let S_i denote the set of nonzero parameters in the i th equation of the VAR system, s_i its cardinality, and $\phi_{\min}^{(i)}$ the minimum nonzero entry (in absolute value) of $\boldsymbol{\Phi}^{*(i)}$. Additionally, define $\sigma_y^{(i)}$ and $\sigma_\varepsilon^{(i)}$ as the standard deviations of $y_t^{(i)}$ and $\varepsilon_t^{(i)}$, and let κ_i represent the restricted eigenvalue of the population covariance matrix $\Gamma = \mathbb{E}(\mathbf{Z}_i \mathbf{Z}_i')$. It is assumed that κ_i is strictly positive, ensuring the sample Gram matrix $\boldsymbol{\Psi}_T = \mathbf{Z}'\mathbf{Z}/T$ approximates Γ .³

Let $\sigma_T = \max_{1 \leq i \leq N} (\sigma_y^{(i)} \vee \sigma_\varepsilon^{(i)})$, $\lambda_T = \sqrt{8 \ln(1+T)^5 \ln(1+N)^4 \ln(1+p)^2 \ln(N^2 p) \sigma_T^4 / T}$, and $0 < q < 1$. Kock and Callot (2015) show that for all $i = 1, \dots, N$,

with a probability of at least $p^* = 1 - 2(N^2 p)^{1 - \ln(1+T)} - 2(1+T)^{-1/4} - \pi_q(s_i)$, we have

$$\frac{1}{T} \left\| \mathbf{Z} \hat{\Phi}^{(i)} - \mathbf{Z} \Phi^{*(i)} \right\|_{\ell_2}^2 \leq \frac{16}{q \kappa_i^2} s_i \lambda_T^2, \quad (9)$$

$$\left\| \hat{\Phi}^{(i)} - \Phi^{*(i)} \right\|_{\ell_1} \leq \frac{16}{q \kappa_i^2} s_i \lambda_T, \quad (10)$$

where $\pi_q(s_i)$ is a constant depending on q and s_i and A is a positive constant. Moreover, with at least probability p^* , no relevant variables will be excluded from the equation if

$$\phi_{\min}^{(i)} > \frac{16}{q \kappa_i^2} s_i \lambda_T^2.$$

Inequality (9) provides an upper bound on the prediction error, while inequality (10) presents an upper bound for the parameter estimation error. The error bounds are influenced by the value of the restricted eigenvalue κ_i : The farther it is from zero, the smaller the upper bounds become. On the other hand, the bounds increase with denser connections, represented by a larger s_i value. For a variable to be selected by LASSO with the given probability, the minimum strength of the connection $\phi_{\min}^{(i)}$ must exceed $16s_i \lambda_T^2 / (q \kappa_i^2)$. The estimation and prediction consistency are achieved if $\lambda_T \rightarrow 0$ as $T \rightarrow \infty$.

The effectiveness of LASSO in variable selection depends on the careful choice of its tuning parameter, λ , which directly impacts the number of variables included in the model in finite sample. In this study, we adopt the coordinate descent algorithm (Friedman et al. 2010) and employ K -fold cross-validation to select the optimal penalty parameters. Specifically, considering Equation (7) as an example, we partition the data set $[\mathbf{y}^{(i)}, \mathbf{Z}]$ into K disjoint subsets along the time dimension, denoted as $J = [J_1, J_2, \dots, J_K]$. During each cross-validation run, we use one of these subsets, say J_k (with $k = 1, 2, \dots, K$), as the validation set and the remaining subsets, that is, $J_{k-} = \cup_{j \neq k} J_j$, for model fitting. Let Λ be the sets of all possible values of λ . For each $\lambda \in \Lambda$, the prediction error over the test set J_k is

$$\text{PE}(\lambda, J_k) = \left\| \mathbf{y}^{(i)} - \mathbf{I} \hat{\phi}_0^{(i)}(\lambda, J_{k-}) - \mathbf{Z} \hat{\Phi}_\lambda^{(i)}(\lambda, J_{k-}) \right\|_{\ell_2}^2. \quad (11)$$

In this equation, $\hat{\phi}_0^{(i)}(\lambda, J_{k-})$ and $\hat{\Phi}_\lambda^{(i)}(\lambda, J_{k-})$ are the LASSO estimators obtained from set J_{k-} with penalty parameter λ , while the ℓ_2 norm is computed for the subset J_k . The K -fold cross-validation error for each $\lambda \in \Lambda$ is then computed as

$$\text{CV}(\lambda) = \sum_{k=1}^K \text{PE}(\lambda, J_k), \quad (12)$$

and the tuning parameter λ is solved by

$$\hat{\lambda} = \underset{\lambda \in \Lambda}{\text{argmin}} \quad \text{CV}(\lambda). \quad (13)$$

There has been a debate on the applicability of cross-validation to dependent time series data, as the procedure assumes

independence among observations (Kock and Callot 2015; Kock et al. 2020; Nicholson et al. 2017). However, recent studies have shown that K -fold cross-validation can be effective for autoregressive models, provided that the residuals are not highly correlated (Bergmeir et al. 2018). As a result, it has been widely used in empirical studies (Nardi and Rinaldo 2011; Milunovich 2020a; Panagiotelis et al. 2019). To select the penalty parameter λ in the LASSO implementation, we used the `glmnet` package (Qian et al. 2013) and performed 10-fold cross-validation (i.e., $K = 10$).

2.3 | Generating and Evaluating Forecasts

In this study, we adopt the direct forecasting method to compute the h -step-ahead predictions for all models. Direct forecasts are based on a horizon-specific estimation model, with the dependent variable being the h -step-ahead forecast value. Specifically, the h -step-ahead direct forecast of VAR(p) model⁴ is given as

$$\hat{\mathbf{y}}_{t+h} = \hat{\Phi}_0(h) + \sum_{j=1}^p \hat{\Phi}_j(h) \mathbf{y}_{t-j+1}, \quad (14)$$

where $\hat{\mathbf{y}}_{t+h} = [\hat{y}_{t+h}^{(1)}, \hat{y}_{t+h}^{(2)}, \dots, \hat{y}_{t+h}^{(N)}]'$ denotes the N -dimensional h -step-ahead forecasts vector, $\hat{\Phi}_0(h)$ and $\hat{\Phi}_j(h)$ are parameter estimates from LASSO regression.

Thus, to generate an h -step-ahead out-of-sample forecast at time period t , the parameter matrices $\Phi_0(h), \Phi_1(h), \dots, \Phi_p(h)$ of Equation (14) are first estimated using n most recent observations spanning the interval $[t-n+1, t]$. Following this estimation, a forecast $\hat{\mathbf{y}}_{t+h}$ is calculated by substituting the values of the lagged variables on the right-hand side of the equation, which are known at time t . For example, in generating a one-step-ahead forecast ($h = 1$) using a VAR(1) model ($p = 1$), the prediction would be as follows:

$$\hat{\mathbf{y}}_{t+1} = \hat{\Phi}_0(1) + \hat{\Phi}_1(1) \mathbf{y}_t. \quad (15)$$

An alternative forecasting technique often encountered in the literature is the recursive approach where forecasts are generated by iteratively applying a one-step-ahead model for h periods. While direct forecasts are generally considered to be more robust to misspecifications, the evidence regarding which method is superior is mixed (Marcellino et al. 2006; Pesaran et al. 2011; Sorjamaa et al. 2007). We favor the direct forecasting approach because it eliminates the need to forecast the exogenous variables used in our empirical analysis.

When comparing forecasts from multiple models, their rankings are determined by the loss function, which measures the distance between the forecasted value and the actual value. In this paper, we use the squared forecast error (SFE) defined as

$$\text{SFE: } L_{t+h}^{(i)} = \left(\hat{y}_{t+h}^{(i)} - y_{t+h}^{(i)} \right)^2. \quad (16)$$

Ranking the computed loss measures can help distinguish between competing models. However, the rankings may vary depending on the data sets used, and hence, it is important to

investigate the statistical significance of the loss differentials. Multiple statistical tests have been developed in the literature in the last three decades. The Diebold and Mariano (1995, DM) test has incontestably been the most commonly used among many early research works, which provides an approach to test for equal predictive ability (EPA). It has been applied widely to compare the accuracy of forecasting models, even though the DM test was not originally designed for model comparison (Diebold 2015). Several extensions and modifications centered around the DM test have been introduced later on (West 1996; Harvey et al. 1998; Clark and McCracken 2001, 2009; Giacomini and White 2006; White 2000; Hansen 2005).

This paper employs the MCS procedure developed by Hansen et al. (2011) to evaluate the performance of multiple forecasting models. The MCS approach provides a set of models that are deemed to be statistically equivalent in terms of forecasting performance, at a given level of confidence. This is achieved by comparing standardized loss differentials, that is, loss differences scaled by their sampling variability, between pairs of models through a sequence of tests. In essence, the MCS allows for the identification of best performing models within a set, while also acknowledging the uncertainty associated with model selection. Given a full set of competing models $\mathcal{M} = [\mathcal{M}_1, \dots, \mathcal{M}_m]$, the pair-wise loss differentials between models $\mathcal{M}_j$ and $\mathcal{M}_k$, $j \neq k$, for forecasting $y_{t+h}^{(i)}$ are calculated as

$$d_{t+h, jk}^{(i)} = L_{t+h, \mathcal{M}_j}^{(i)} - L_{t+h, \mathcal{M}_k}^{(i)}. \quad (17)$$

The null hypothesis of EPA is given by

$$H_0: \mathbb{E} \left[d_{t+h, jk}^{(i)} \right] = 0, \quad \forall \mathcal{M}_j, \mathcal{M}_k \in \mathcal{M} \text{ and } j \neq k. \quad (18)$$

If H_0 is rejected at the significance level α using the T_{max} statistic of Hansen et al. (2011), the underperforming model will be removed and the process will continue until nonrejection occurs, leading to a set of surviving models that are statistically equivalent.

2.4 | Robustness Analysis

We evaluate the robustness of our study by incorporating exogenous variables, using alternative evaluation metrics and applying a alternative strategy for constructing MCSs.

2.4.1 | Accounting for Exogenous Macroeconomic Variables

First, we consider the inclusion of macroeconomic variables in the forecasting models. The univariate AR model with exogenous variables is commonly referred as the “ARX” model, while the multivariate VAR specification that includes exogenous variables is known as the “VARX” model. We can easily extend the AR equation for each variable given in (1) as follows:

$$y_t^{(i)} = \phi_0^{(i)} + \sum_{j=1}^p \phi_j^{(i)} y_{t-j}^{(i)} + \sum_{s=1}^k \Lambda_s^{(i)} \mathbf{x}_{t-s} + \varepsilon_t^{(i)}, \quad (19)$$

where $\mathbf{x}_t$ is an M -dimensional vector of exogenous variables at time t and $\Lambda_s^{(i)}$ are the corresponding parameter vectors. The extended model is referred to as ARX(p, k). If we incorporate these exogenous variables into the VAR(p) model in Equation (4), we can similarly obtain the VARX(p, k) model, which is expressed as follows:

$$\mathbf{y}_t = \Phi_0 + \sum_{j=1}^p \Phi_j \mathbf{y}_{t-j} + \sum_{s=1}^k \Lambda_s \mathbf{x}_{t-s} + \varepsilon_t. \quad (20)$$

2.4.2 | Absolute Forecast Error

It is well-known that the SFE loss, as defined in (16), is sensitive to outliers. An alternative loss function which weighs small and large forecast errors equally is the absolute forecast error (AFE) given by

$$\text{AFE}: L_{t+h}^{(i)} = \left| \hat{y}_{t+h}^{(i)} - y_{t+h}^{(i)} \right|. \quad (21)$$

2.4.3 | Conditional Superior Predictive Ability (CSPA)

The CSPA test, as outlined by Li et al. (2022), posits that the benchmark predictive model's conditional expected loss should be as good as or better than that of any rival models across all conditioning states. This test accounts for heterogeneity across sample periods, such as housing market expansion or collapsing, thereby taking the state of the market into consideration when evaluating forecasting performances. The null hypothesis of the CSPA test involves the loss differential $d_{t+h, jk}^{(i)}$ between the benchmark model k and the competing model j defined in Equation (17) as follows:

$$H_0: \mathbb{E} \left[d_{t+h, jk}^{(i)} \mid \mathcal{W}_{t+h} \right] \geq 0, \quad (22)$$

where $\mathcal{W}_{t+h}$ is a conditioning variable. We form the confidence set for the most superior (CSMS) by rotating the benchmark role across all models, comprising all nonrejected benchmarks.

2.5 | Simulation Study

We perform simulation experiments calibrated to the results of our empirical study in order to explore the impact of different levels of system sparsity and connectivity strength on forecast accuracy. Our analysis utilizes a VAR(1) framework for its simplicity and also due to its effectiveness on our actual data. We formulate our Data Generating Process (DGP) as follows:

$$\text{DGP: } \mathbf{y}_t = \Phi_0 + \Phi_1 \mathbf{y}_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \Sigma_\varepsilon). \quad (23)$$

We assume that ε_t is a martingale difference sequence, and all eigenvalues of Φ_1 have modulus less than one. These assumptions result in a weakly stationary process and are commonly found in similar studies. The matrix Φ_0 is set equal to the intercept obtained from a VAR(1) model estimated for the first 30 LGAs (in the order listed in Appendix A2) within the Sydney region from August 2005 to February 2022. Σ_ε is diagonal and contains values obtained from the same estimation.

The elements of Φ_1 control the levels of sparsity and connectivity strength and are also calibrated to the results of our empirical results as detailed in Section 5.1.

Having described the DGP, we next enumerate the steps of our simulation study. As can be seen, Step 1 involves generating the data from the DGP, while Steps 2 and 3 closely mimic our empirical methodology to generate the forecasts and compute MCSs. Finally, Step 4 averages the results across all replications, allowing us to assess the effectiveness of the competing models on the synthetic data.

Step 1: Simulate the dataset $\mathbf{y}_t^{(b)}$, $b = 1, \dots, B$ using the DGP from (23) for each considered value of Φ_1 . The number of simulations is set to $B = 300$.

Step 2: Calculate the h -step-ahead out-of-sample predictions for $\mathbf{y}_t^{(b)}$ using a number of different forecasting models. The forecasts are computed based on a rolling window forecasting scheme with the estimation window size t_0 set to 70% of the dataset.

Step 3: Let $\hat{\mathbf{y}}_{t+h}^{(i,b,m)}$, $t = t_0, \dots, T$, be the forecast from model m for the i th variable of $\mathbf{y}_{t+h}^{(b)}$. We first compute the forecast loss $L_{t+h}^{(i,b,m)}$ for all models and then implement the MCS test in order to obtain a set of “superior” models at a given confidence level, which we set to 90%.

Step 4: Given the MCS test results for all the simulated data, we compute the average proportion of time that a model m has been included in a MCS across all variables and replications

$$\bar{f}_m = \frac{1}{B} \sum_{b=1}^B f_m^{(b)} \quad \text{with} \quad f_m^{(b)} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}_m^{(i,b)}, \quad (24)$$

where

$$\mathbb{I}_m^{(i,b)} = \begin{cases} 1, & \text{if model } m \text{ is in a MCS for the } i \text{ th variable in the } b \text{ th replication} \\ 0, & \text{otherwise.} \end{cases}$$

3 | Data and Empirical Study

In this section, we forecast disaggregated house price growth rates from three housing markets, namely, within-city house price growth rate for Sydney and Melbourne (Australia) and city-level house price growth rates from China. We evaluate out-of-sample predictions generated by our LDVAR models, which incorporate price interconnectedness, and compare their predictive performance against univariate models which ignore regional linkages.

3.1 | Data

For the Australian markets, we use monthly house prices at the LGA level for Sydney and Melbourne from August 2005 to February 2022. As the two largest cities in Australia and the respective capitals of New South Wales and Victoria, Sydney and

Melbourne represent the most prominent housing markets in the Australian economy. The LGAs, which encompass areas greater than a single suburb but smaller than a capital city, constitute the lowest tier of government in Australia. Consistent with Hurn et al. (2022), our analysis encompasses 32 LGAs for Sydney and 31 LGAs for Melbourne, which are listed in Table A2. The house price data are obtained from the CoreLogic RP database of the Securities Industry Research Centre of Asia-Pacific (SIRCA) and are adjusted for inflation using the CPI (excluding shelter) obtained from the Australian Bureau of Statistics.

Regarding the Chinese market, we use monthly city-level house price indices which have been compiled by Fang et al. (2016). The sample period used in the empirical analysis is January 2003 to March 2013. The conversion of nominal prices to real prices is performed using the city-level urban consumer price indices obtained from the China Statistical Yearbook. A comprehensive list of the 88 cities included in this study, comprising of 4 Tier 1, 26 Tier 2, and 58 Tier 3 cities, is presented in Table A3 of Appendix A2.

While our housing price series are available monthly, the CPI data we collected are at the quarterly frequency. To synchronize these series and compute real house prices, we convert both the Australian and Chinese price indices from quarterly to monthly frequency via linear interpolation as in Hurn et al. (2022) and Sekine and Tsuruga (2018). Thus, our subsequent analysis is conducted on real house price series that are derived by this method. Figure 1 depicts the natural logarithm of real house prices in the Australian and Chinese markets, accompanied by the average autocorrelation function (ACF) for the log growth rates. We make three key observations here.

First, as evident by the time series plots for the LGA house price indices in Sydney and Melbourne, there are strong cyclical comovements within these two cities. Given the narrower spread (variability) in the first plot, it appears that the interconnectedness is somewhat stronger within the Sydney region. Concerning the plot for the city-level house prices in the Chinese market, we observe an overall upward pattern over the time period of investigation but also a substantial amount of noise.

Second, we note that our sample period includes several crises that could have led to instability in the data-generating process (DGP). For instance, the Australian data sample extends from August 2005 to February 2022. As such, it includes the Global Financial Crisis (2007–2008), the European Debt Crisis (2010–2013), and the recent COVID-19 pandemic period (2020). Some of the effects of those events are observable in Figure 1, where they correspond with the house prices in Sydney and Melbourne exhibiting local troughs and a tighter grouping of the log price series. To account for this observation, we apply the rolling window estimation technique which excludes relatively “old” data points from the estimation dataset and uses only the most recent subset of observations to estimate the parameters. This method helps prevent the contamination of model forecasts with outdated information, thereby reducing the likelihood of biased forecasts.⁵

Our third observation is regarding the average ACFs computed for the house price growth rates and presented in the second column of Figure 1. Here, we note substantially more

Logarithm of House Prices vs. Averaged ACF of Price Returns

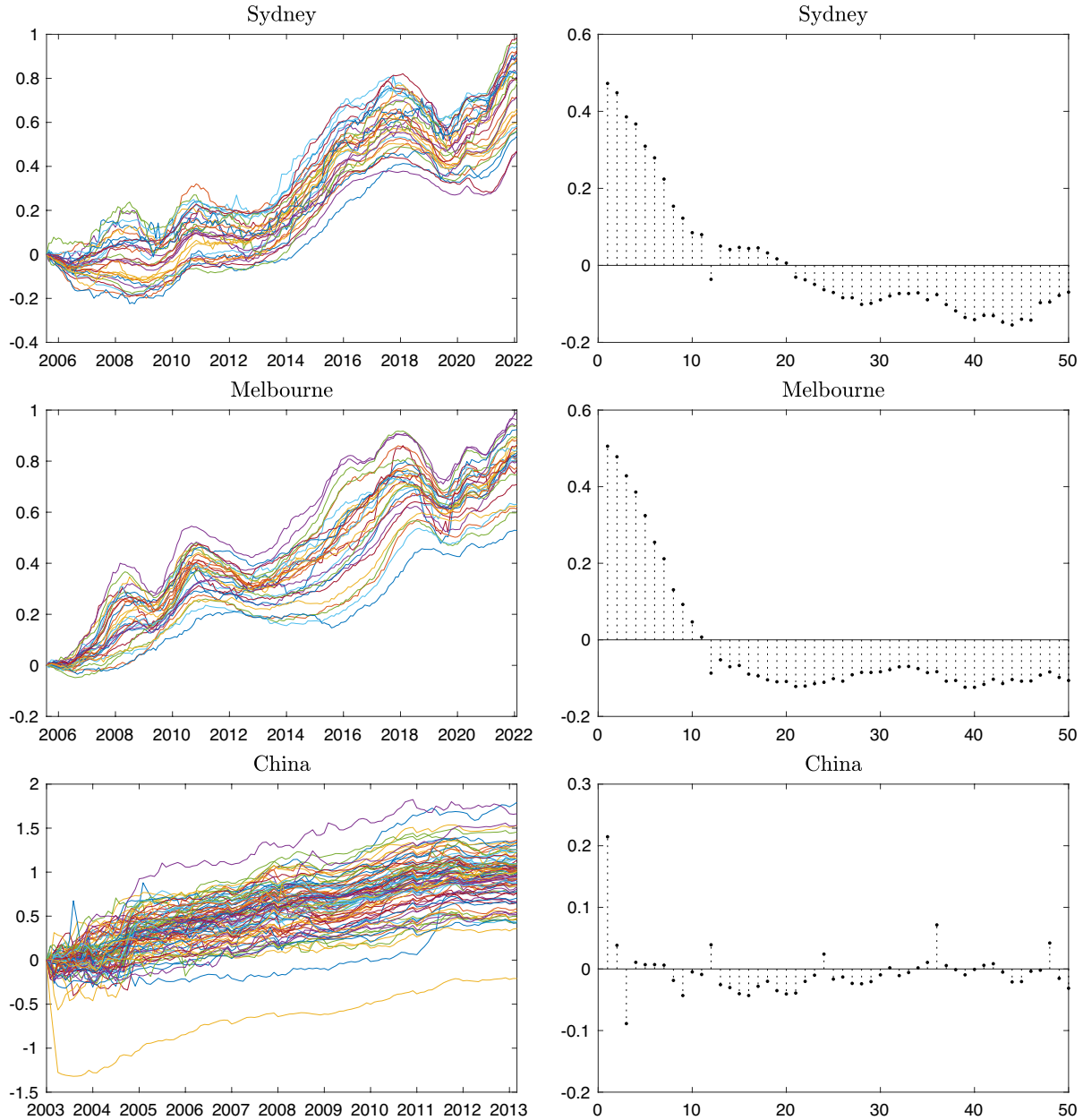


FIGURE 1 | The left panel shows the time series plot of logarithmic real monthly house prices for the Australian and Chinese markets, spanning the period August 2005 to February 2022 for the Australian market and January 2003 to March 2013 for the Chinese market. The right panel displays the averaged ACF computed on log growth rates (returns). The analysis includes data from 32 LGAs in Sydney, 31 LGAs in Melbourne, and 88 cities in China, with each line representing one LGA or city.

persistence in the monthly data for Sydney and Melbourne LGAs relative to the Chinese data. This is indicative of more complex and richer dynamics in the house return returns for the LGAs in Sydney and Melbourne, relative to the series for the Chinese cities.

Based on the above discussion, we can hypothesize that the LGA housing markets within Sydney and Melbourne exhibit more connections (i.e., less sparsity) which are of greater magnitude (i.e., connectivity strength) relative to the Chinese city-level markets. This hypothesis is partly supported by the findings of Hurn et al. (2022), which indicate that by 2017, approximately 50% of all possible interconnections within both

the Sydney and Melbourne housing markets were active. In contrast, the Chinese housing market had only 20% active connections in 2013—significantly lower to the figures recorded in Australia. These differences may impact the accuracy and usefulness of the multivariate LDVAR models used in our subsequent analysis.

3.2 | Results: Out-of-Sample Forecasting

In this section, we consider price connectedness from two perspectives. First, we evaluate whether incorporating connectedness through LDVAR models enhances house price forecasts

compared to univariate models that do not account for connectedness. Second, we investigate the degree of connectedness captured in the estimated models for each of the three housing markets under investigation. We treat each of the three housing markets as a separate system and perform rolling window estimation of the LDVAR model by applying the LASSO method, as outlined in Section 2.2. In addition to three LDVAR models (with 1, 2, and 3 lag terms, respectively), we also produce forecasts from three AR specifications and the ARMA(1,1) model. These univariate models do not account for the interconnect- edness of house prices and are used to evaluate whether incor- porating this interconnectedness can improve out-of-sample forecasts.

The estimation (training) of the models is performed on 70% of the available data, which translates into a window size of 138 observations for Sydney and Melbourne markets and 85 data points for the Chinese markets. Forecasts are then constructed for 1-, 3-, 6- and 12-month ahead periods and evaluated on in- dependent test data. This provides us with (i) sixty 1-month ahead forecasts for each LGA in the Sydney and Melbourne markets and 37 predictions for each city in China, (ii) 58 and 35 respective forecasts for 3-month ahead, (iii) 55 and 32 predic- tions for 6-month ahead predictions, and finally, (iv) forty-nine and twenty-six 12-month ahead forecasts, respectively. For each rolling window estimation, out-of-sample forecast errors are computed and the MCS procedure performed, that is, see (18) to obtain the best model set for each LGA or city at the given forecasting horizon. We then calculate the frequency that a par- ticular model is in the set of best performing models for each market and present these figures in Table 1. Note that the MCS may include several models so that the sum of the frequencies across each row in Table 1 can easily exceed one.

Focusing on the Sydney and Melbourne markets, multivariate models clearly outperform the univariate specifications at all forecasting horizons. The only exception is in the case of the 12-month ahead forecast for Melbourne when the AR(1) slightly outperforms. The VAR(1) model performs particularly well at 1-month ahead forecasting and is present in all computed MCSs, as indicated by the selection frequency of 100%. In contrast, AR(1) which is the simplest of the models applied here, is in the MCS only 43.8% and 29.0% of the time for the Sydney and Melbourne markets.

For the Chinese market, the AR(1) specification performs better at all horizons except the 12-month ahead forecast, when the VAR(1) model marginally outperforms (63.6% versus 61.4%). These findings are in line with the initial observations about the time series plots for the three markets provided above. These results are also consistent with our simulations discussed in Section 5, where we report that LDVAR models forecast better in relatively dense systems with strong connectivity, while uni- variate models tend to outperform in sparse systems with weak connectivity strengths.

After evaluating how accounting for connectedness impacts forecasting performance in Table 1, we now turn our atten- tion to assess the degree of connectedness in the estimated equations. As previously discussed, we characterize con- nectedness with the sparsity and connectivity strength of the lagged parameter matrices. Figures 2 and 3 summarize this information by showing the average selection frequencies for each regional unit within the three housing markets and the magnitudes of the estimated parameters. These values are cal- culated from VAR(1) models used for one-step-ahead predic- tions, as described in (15).

TABLE 1 | Out-of-sample results for the Australian and Chinese housing markets at various forecasting horizons based on 90% MCS test results.

	AR(1)	AR(2)	AR(3)	ARMA(1, 1)	VAR(1)	VAR(2)	VAR(3)
Sydney							
$h = 1$	43.8	50.0	40.6	50.0	100.0	81.3	78.1
$h = 3$	68.8	78.1	78.1	59.4	87.5	87.5	87.5
$h = 6$	65.6	56.3	56.3	78.1	84.4	56.3	56.3
$h = 12$	68.8	59.4	40.6	78.1	75.0	81.3	71.9
Melbourne							
$h = 1$	29.0	38.7	38.7	38.7	100.0	58.1	61.3
$h = 3$	71.0	77.4	71.0	51.6	80.6	87.1	93.5
$h = 6$	77.4	67.7	61.3	77.4	93.5	90.3	71.0
$h = 12$	83.9	80.6	80.6	77.4	67.7	80.6	80.6
China							
$h = 1$	83.0	62.5	51.1	63.6	55.7	44.3	42.0
$h = 3$	75.0	65.9	62.5	64.8	60.2	54.5	58.0
$h = 6$	73.9	61.4	56.8	65.9	61.4	52.3	53.4
$h = 12$	61.4	56.8	44.3	60.2	63.6	55.7	55.7

Note: The table displays the percentage of inclusion of each competing model in the MCSs, which are based on the SFE loss function.

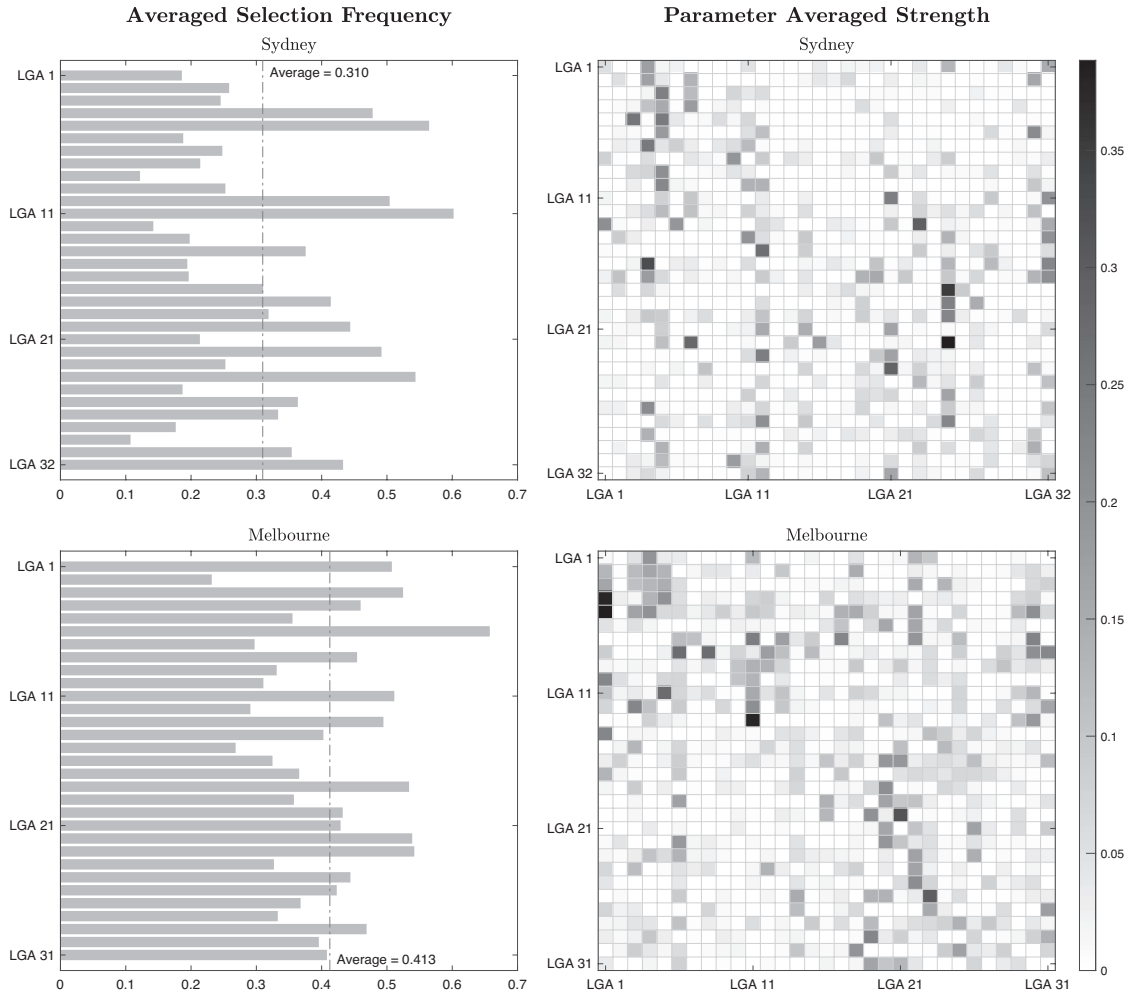


FIGURE 2 | Average connection frequency and average parameter strength in the VAR(1) model ($h = 1$) for the Australian markets. The figure displays the average selection frequency and the heat map of the average estimated parameter strength across the rolling window for the Australian market in the VAR(1) model for the 1-month ahead forecast.

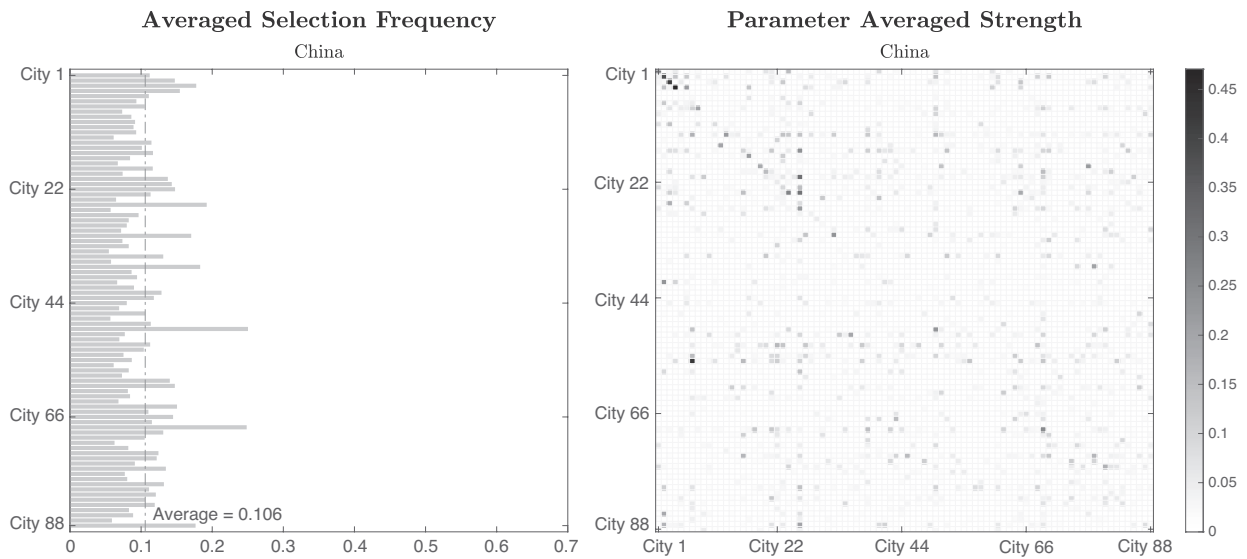


FIGURE 3 | Averaged parameter connection frequency and averaged parameter strength in the VAR(1) model ($h = 1$) for the Chinese market. The figure displays the averaged selection frequency and the heat map of the average estimated parameter strength across the rolling window for the Chinese market in the VAR(1) model for the one-step-ahead forecast.

Average selection frequencies represent the proportion of time an LGA/city plays a role in forecasting 1-month ahead house price in all other LGAs/cities within their region. They are calculated as the frequencies of relevant nonzero coefficients estimated by the LASSO technique and are indicative of the level of sparsity in the estimated parameter matrix. The averaged percentages (across all rolling windows) are displayed in the left panel of Figures 2 and 3. In general, all LGAs/cities have been selected as relevant predictors by LASSO, albeit some areas are selected more often than others. The averaged selection frequencies range between 10.7% and 60.2% for different LGAs in Sydney, 23.1% and 65.8% for Melbourne, and 5% and 25% for the cities in China. The higher the average selection frequency, the lower the sparsity of the lagged parameter matrix. Thus, the housing markets in Sydney and Melbourne appear to be more densely connected compared to the Chinese markets, with the mean link being 31.0% (41.3%) for Sydney (Melbourne) and only 10.6% for China. This result is consistent with the evidence provided in Hurn et al. (2022), who use a different method to conclude that within-city housing networks in Sydney and Melbourne are more interconnected than those found in the cross-city housing markets of China.

Plots illustrating the magnitudes of the estimated lagged coefficients, which indicate our second measure of connectedness—connectivity strength—are provided in the right columns of Figures 2 and 3. These values are calculated by averaging the estimated VAR parameters for each pair of LGAs (cities) across the rolling window estimations, resulting in total 726 and 793 links in Sydney and Melbourne, respectively, and 4251 connection links for China. The computed average strengths range between 0.00 and 0.32 for Sydney, 0.00 and 0.37 for Melbourne, and between 0.00 and 0.45 for China. The minimum and maximum average connection strengths discussed above, however, do not fully capture the distribution of these strengths within each market. To gain a better understanding of the estimated connectivity strengths, it is useful to calculate the frequencies with which the estimated linkages fall into weak, moderate, and strong ranges. For this purpose, we define the weak range of connectedness to be the $[-0.05, 0.05]$ interval and find that 75.9%, 71.6%, and 92.1% of the links for Sydney, Melbourne, and China, respectively, are weak. In the moderate range of links, as given by the parameter interval $[-0.15, -0.05] \cup [0.05, 0.15]$, we find 17.5%, 21.7%, and 7.3% of connections for Sydney, Melbourne, and China, respectively. Finally, in the range of strong connections (less than -0.15 or greater than 0.15), we have 6.6%, 6.7%, and 0.7% of the connections within the same three regions. Thus, we observe that housing markets for Sydney and Melbourne exhibit higher frequencies of moderate and strong connectivity strengths relative to the Chinese housing network.

In summary, our results presented here show that not only Sydney and Melbourne markets are more densely connected (exhibit less sparsity in the lagged LDVAR parameter matrix), but that they also have higher proportions of moderate/strong connections relative to the Chinese markets. This may explain why the LDVAR model is preferred for forecasting in the Sydney and Melbourne LGA markets, while the univariate AR specification is more suitable for predicting house price returns in

Chinese city markets, even though all three markets exhibit a degree of within-market interconnectedness. However, we will defer making this conclusion until we present the results of our simulation study in Section 5.

4 | Robustness Analysis

In the previous section, we provide results derived from a pure LDVAR model, that is, without taking into account any exogenous variables. However, given that house prices are influenced by key macroeconomic factors, it is worth exploring if including such factors can affect our results and how. In this section, we conduct a number of sensitivity checks and demonstrate that our results are robust to (i) the inclusion of macroeconomic variables in the forecasting models, (ii) the use of alternative loss functions, and (iii) the use of an alternative test that considers the prevailing state of the market to evaluate the relative predictive capabilities of various models.

4.1 | Predictive Models With Exogenous Macroeconomic Variables

A number of studies have shown that incorporating exogenous macroeconomic variables such as interest rates and unemployment figures into housing price forecasting models can increase their accuracy; see, for example, Rapach and Strauss (2007) and Bork and Müller (2018). In this section, we expand on both the AR and VAR models by integrating a set of exogenous variables as described in (19) and (20) to assess whether our previously reported findings are sensitive to the introduction of exogenous predictors.

Table 2 presents the lists of exogenous variables included in the models for the two Australian city markets and the Chinese housing market, which are provided in the top and bottom panels, respectively. As shown in the table, we include a variety of both national and regional/local variables that may be useful in predicting disaggregated house price growth rates. Nearly all of the included variables, or their variations, appear in the list of predictors considered in the existing literature, including studies such as Rapach and Strauss (2007, 2009), Gupta (2013), and Shi and Phillips (2023). As noted in the third column, these variables are collected at monthly, quarterly, and annual frequencies. In order to synchronize their values with the house prices, we convert all variables to monthly frequency using linear interpolation. This is further discussed in the Data Section (Section 3.1).

All variables are expressed in real terms, except for the credit supply growth rate in the Australian market and the M2 money supply growth rate in the Chinese market. Regarding these nominal variables, empirical research has now shown that changes in nominal prices can influence and help predict real house prices. Studies that utilize inflation to model real house prices and growth rates include Rapach and Strauss (2007, 2009); Plakandaras et al. (2015) for the US market, Abelson et al. (2005); Milunovich (2020a) for Australia, and Stevenson (2000) for the United Kingdom, among others. In addition to inflation, it is also important to account for the stance of macroprudential regulation in Australia—a policy implemented by the Australian Prudential Regulation

TABLE 2 | List of macroeconomic variables included in models (19) and (20) for the empirical analysis.

Macroeconomic variable	Source	Frequency	Area
Australian market			
Real mortgage rate	RBA	Monthly	National
Credit supply growth Rate	RBA	Monthly	National
10-year to 3-month bond spread	RBA	Monthly	National
Population growth rate	ABS	Quarterly	State
Unemployment growth rate	ABS	Monthly	State
Housing completion growth rate	ABS	Quarterly	State
Real state final demand growth rate	ABS	Quarterly	State
Real house rental growth rate	SIRCA	Monthly	LGA
House listing growth rate	SIRCA	Monthly	LGA
Chinese market			
Real interest rate	FRED database	Monthly	National
M2 money supply growth rate	FRED database	Monthly	National
Real GDP per capital growth rate	China City Statistical Yearbook	Yearly	City
Population growth rate	China City Statistical Yearbook	Yearly	City
Unemployment growth rate	China City Statistical Yearbook	Yearly	City

Note: This table provides a list of the macroeconomic variables used in the empirical analysis of the Australian and Chinese markets. Note that annually and quarterly frequency data are converted to monthly frequency via linear interpolation.

TABLE 3 | MCS results for the $ARX(p, k)$ and $VARX(p, k)$ models, $p = 1, 2, 3$ and $k = 6$, for the Australian and Chinese housing markets.

	ARX(1)	ARX(2)	ARX(3)	VARX(1)	VARX(2)	VARX(3)
Sydney						
$h = 1$	21.9	37.5	40.6	93.8	65.6	59.4
$h = 3$	46.9	71.9	71.9	59.4	59.4	78.1
$h = 6$	65.6	68.8	53.1	68.8	68.8	84.4
$h = 12$	84.4	81.3	59.4	78.1	78.1	87.5
Melbourne						
$h = 1$	25.8	54.8	51.6	90.3	77.4	77.4
$h = 3$	45.2	64.5	67.7	74.2	71.0	93.5
$h = 6$	51.6	51.6	45.2	71.0	96.8	93.5
$h = 12$	87.1	74.2	77.4	74.2	80.6	90.3
China						
$h = 1$	77.3	62.5	51.1	62.5	50.0	59.1
$h = 3$	76.1	60.2	58.0	64.8	77.3	67.0
$h = 6$	70.5	59.1	61.4	69.3	69.3	71.6
$h = 12$	69.3	56.8	47.7	69.3	67.0	63.6

Note: The table displays the percentage of time each model is included in the 90% MCS based on the SFE loss. The maximum value among all the competing models is highlighted in boldface.

Authority (APRA) through adjustments to lending conditions. APRA has actively used credit controls to influence housing market outcomes. For example, in December 2014, APRA

introduced a 10% annual growth cap on investor housing credit for banks, which had a significant impact on housing demand and, consequently, prices. In our predictive models,

the effects of inflation and credit conditions on the housing market are jointly captured through the variable representing the nominal credit supply growth rate.

In this robustness check, we maintain a fixed lag order of $k = 6$ for the exogenous variables in all models. The lag order for the macroeconomic variables is based on the understanding that

TABLE 4 | Out-of-sample forecasting results for the Australian and Chinese housing markets at different forecasting horizons based on 90% MCS test results with the AFE loss function.

	AR(1)	AR(2)	AR(3)	ARMA(1, 1)	VAR(1)	VAR(2)	VAR(3)
Sydney							
$h = 1$	40.6	56.3	50.0	59.4	93.8	71.9	78.1
$h = 3$	53.1	53.1	53.1	34.4	84.4	75.0	65.6
$h = 6$	65.6	65.6	53.1	78.1	84.4	62.5	71.9
$h = 12$	71.9	62.5	53.1	68.8	71.9	78.1	78.1
Melbourne							
$h = 1$	41.9	51.6	38.7	58.1	96.8	77.4	74.2
$h = 3$	64.5	71.0	67.7	54.8	71.0	83.9	90.3
$h = 6$	74.2	61.3	54.8	67.7	90.3	80.6	61.3
$h = 12$	90.3	80.6	67.7	77.4	74.2	83.9	87.1
China							
$h = 1$	88.6	71.6	61.4	62.5	62.5	47.7	50.0
$h = 3$	73.9	64.8	62.5	58.0	64.8	63.6	56.8
$h = 6$	75.0	63.6	55.7	68.2	56.8	64.8	59.1
$h = 12$	70.5	62.5	45.5	64.8	65.9	61.4	56.8

Note: The table displays the percent of the time that each model is included in the 90% MCS under the AFE loss function. The maximum value among all the competing models is highlighted in boldface.

TABLE 5 | Out-of-sample forecasting results for the Australian and Chinese housing markets based on the 90% CSPA test and the SFE loss function.

	AR(1)	AR(2)	AR(3)	ARMA(1, 1)	VAR(1)	VAR(2)	VAR(3)
Sydney							
$h = 1$	40.6	59.4	56.3	50.0	93.8	78.1	81.3
$h = 3$	43.8	68.8	71.9	56.3	71.9	68.8	68.8
$h = 6$	50.0	53.1	46.9	56.3	78.1	43.8	53.1
$h = 12$	25.0	37.5	25.0	50.0	81.3	90.6	68.8
Melbourne							
$h = 1$	25.8	51.6	58.1	64.5	93.5	93.5	96.8
$h = 3$	29.0	58.1	41.9	35.5	71.0	90.3	90.3
$h = 6$	32.3	32.3	29.0	25.8	93.5	80.6	80.6
$h = 12$	54.8	38.7	22.6	41.9	51.6	83.9	90.3
China							
$h = 1$	78.4	67.0	61.4	71.6	69.3	67.0	60.2
$h = 3$	70.5	69.3	69.3	67.0	72.7	72.7	72.7
$h = 6$	75.0	77.3	71.6	70.5	73.9	68.2	65.9
$h = 12$	71.6	69.3	60.2	68.2	70.5	70.5	60.2

Note: The table displays the percentage of the time that each model is included in the 90% CSPA-based most superior confidence set under the SFE loss. The maximum value among all the competing models is highlighted in boldface.

housing markets may react to economic factors with some delay. Here, we follow the literature to incorporate six lags of macro-economic variables which account for the lagged effects of the economic variables for up to two quarters as in Dungey and Pagan (2009) and Robstad (2018).

Following the inclusion of the exogenous variables in the expanded information set, we recompute the MCS and provide a ranking of our models. Table 3 shows that the MCS results are broadly consistent with those from Table 1, where we do not include any exogenous variables. For the Sydney and Melbourne markets, multivariate models outperform univariate models

at all forecasting horizons. For the Chinese city markets, the ARX(1) model performs better in the immediate 1-month ahead and ranks equally with VARX(1) at the 12-month ahead forecast. The multivariate models perform better at the 3- and 6-month ahead forecasts.

4.2 | Absolute Forecast Errors

Here we re-evaluate real housing returns predictions using the AFE loss provided in (21), rerank the forecasting models accordingly, and present our new results in Table 4 below.

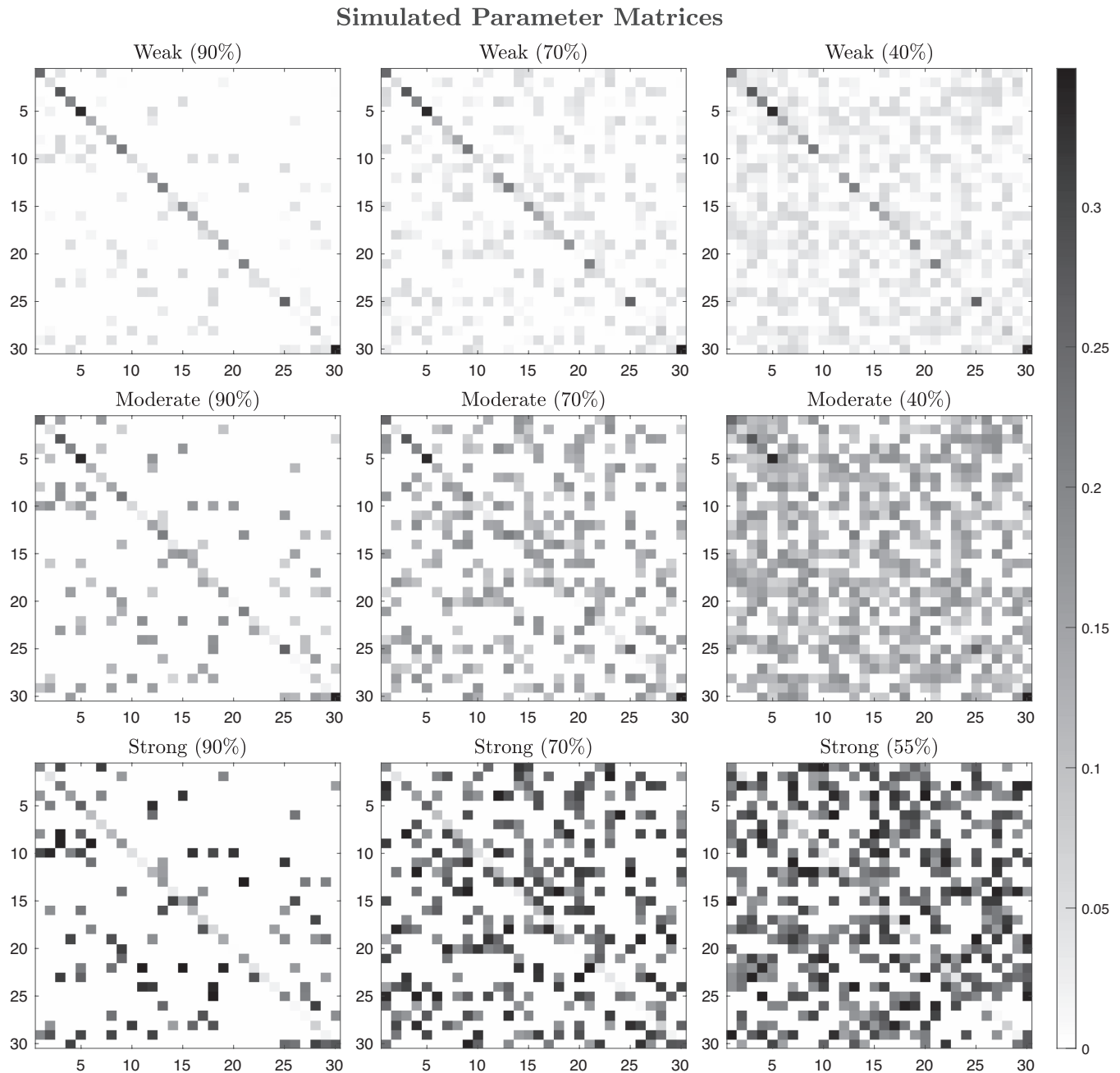


FIGURE 4 | The heat map of parameter matrices for the VAR(1) model constructed under different connection strengths and sparsity levels. The connection strengths are classified into three categories: Weak, Moderate, and Strong where the degree of shading represents the magnitude of the coefficients, with darker colors indicating larger magnitudes. The sparsity levels range from 90% to 40%.

Overall, we observe a pattern that is similar to what we have in Table 1. Namely, multivariate LDVAR models appear most appropriate for forecasting LGA house price returns in the Sydney and Melbourne markets, while for the Chinese city-level series, the univariate AR(1) specification is included in the MCS most often. Additionally, there is a decrease in the frequency of LDVAR models being selected in the MCS, relative to the SFE results; however, this reduction is only marginal. The only exception is the 12-month ahead forecast for the Chinese housing markets where there is a reversal in the best ranking model in favor of AR(1). To summarize, we can conclude that our model rankings are relatively robust to the application of alternative loss functions as demonstrated using the AFE loss.

4.3 | CSPA Test

Next, we examine the sensitivity of our results to the MCS procedure by employing a different statistical test that evaluates the predictive capabilities of our models based on conditional expected loss, as outlined in Equation (22). The chosen conditioning variables for the Sydney and Melbourne markets are the first principal components of the price–rent ratios across their respective LGAs, which reflect the overall states of each market. Since the rent data are not available for the Chinese market, we use income data to replace the rent, leading to the price–income ratio. Thus, the conditioning variable for the Chinese market is the first principal component of the price–income ratio. Those measures are often used to evaluate housing affordability and have been employed to identify housing bubbles (see, for example, Shi et al. 2020; Chen et al. 2022; Shi and Phillips 2023).

Table 5 shows the forecasting performance using the CSPA test. In general, our results for the Sydney and Melbourne housing markets remain unchanged, except that AR(3) in Sydney is now as good as the VAR(1) model for the 3-month ahead forecast. Concerning the Chinese market, our results are again qualitatively consistent with our previous findings with the univariate models outperforming the multivariate specifications, except at the 3-month ahead forecasting horizon.

5 | Simulation Study

Given our empirical results, it remains uncertain what conditions of within-market connectedness, as captured by LDVAR systems, lead to improved forecast accuracy compared to univariate models. The LASSO estimation technique functions as a variable selection device, eliminating zero-value coefficients and retaining only those parameters that are nonzero. As nonzero coefficients represent interrelationships among variables, LASSO estimation is an effective method for identifying such connections in a large-dimensional setting. Yet, the upper bound of the forecast error of the VAR system increases with the number of nonzero coefficients and diminishes with restricted eigenvalue, as given in Equation (9). In this section, we explore how different levels of sparsity and connectivity strength impact out-of-sample prediction accuracy by conducting simulation experiments under a range of connectivity scenarios.

5.1 | Generating the Data

We simulate data using the model described in Equation (23). The dimension of the multiple time series $\mathbf{y}_t$ is set to 30, that is, $N = 30$, which is close to the number of LGAs in Sydney and Melbourne. The length of the simulated time series is set to 250, that is, $T = 250$, equivalent to approximately 20 years of monthly data. Given that this simulation study explores the effects of sparsity and connectivity strength on the forecasting performance, special attention needs to be given to the construction of the parameter matrix Φ_1 . Sparsity pertains to the proportion of zero parameters in a matrix; that is, a sparsity of 90% implies that 90% of all matrix elements are zeros, whereas connectivity strength relates to the magnitude of the off-diagonal parameters. To investigate these effects, we construct nine parameter matrices with varying levels of sparsity and strength.

We explore three tiers of connectivity strength—weak, moderate, and strong—each accompanied by varying degrees of sparsity (90%, 70%, and 40% for weak and moderate and 90%, 70%, and 55% for strong).⁶ Since the diagonal elements of Φ_1 do not play a role in our definition of connectivity

TABLE 6 | Summary statistics of simulated datasets under varying strength and sparsity settings.

Strength	Sparsity	μ	σ	$\max(\psi_T)$	L_1	L_5	L_{10}
Weak	90%	0.001	0.009	0.275	0.012	-0.005	-0.004
	70%	0.001	0.009	0.269	0.011	-0.005	-0.004
	40%	0.001	0.009	0.287	0.011	-0.004	-0.004
Moderate	90%	0.001	0.009	0.304	0.013	-0.005	-0.004
	70%	0.001	0.010	0.402	0.012	-0.004	-0.004
	40%	0.001	0.010	0.512	0.016	-0.001	-0.004
Strong	90%	0.001	0.010	0.523	0.022	-0.002	-0.005
	70%	0.000	0.014	0.880	0.071	0.037	0.018
	55%	0.001	0.023	0.951	0.079	-0.015	0.058

Note: The table displays the mean (μ), standard deviation (σ), and the maximum eigenvalue ($\max(\psi_T)$) of Φ_1 for the simulated data. Additionally, the averaged autocorrelation function (ACF) at Lags 1, 5, and 10 is also reported.

TABLE 7 | The averaged percentage of one model being included in a 90% MCS at different forecasting horizons, built on the SFE loss function.

		Univariate		Multivariate	
Strength	Sparsity	AR(1)	ARMA(1,1)	VAR(1)	VAR(2)
$h = 1$					
Weak	90%	86.8	77.4	75.5	74.4
	70%	87.3	77.7	77.1	74.9
	40%	87.5	77.6	77.5	74.6
Moderate	90%	84.4	75.0	79.9	74.7
	70%	77.9	69.3	86.8	74.6
	40%	60.6	55.2	93.9	71.7
Strong	90%	37.5	33.5	93.6	74.7
	70%	0.9	0.8	97.6	59.4
	55%	0.0	0.0	99.5	35.3
$h = 3$					
Weak	90%	83.5	77.4	83.2	82.0
	70%	82.4	77.6	84.7	82.8
	40%	83.5	77.6	83.9	82.7
Moderate	90%	83.3	77.8	83.5	82.1
	70%	82.5	77.4	83.7	82.4
	40%	83.0	77.5	83.5	81.4
Strong	90%	83.0	77.2	84.0	81.7
	70%	56.9	53.8	91.8	76.3
	55%	2.5	2.4	98.3	55.8
$h = 6$					
Weak	90%	83.3	78.5	82.6	81.3
	70%	82.6	78.8	83.2	81.7
	40%	82.9	77.9	83.7	82.0
Moderate	90%	82.5	78.3	81.8	81.0
	70%	82.8	79.0	83.4	82.7
	40%	83.9	78.8	84.4	82.3
Strong	90%	83.0	78.4	82.8	82.2
	70%	81.0	76.8	83.3	80.4
	55%	37.6	36.1	95.3	72.2
$h = 12$					
Weak	90%	84.5	79.4	82.4	80.8
	70%	83.3	80.1	82.5	81.0
	40%	82.9	79.4	82.7	81.1
Moderate	90%	83.9	79.4	82.6	81.4
	70%	83.5	79.2	82.2	81.3
	40%	82.9	79.4	82.8	81.3

(Continues)

TABLE 7 | (Continued)

Strength	Sparsity	Univariate		Multivariate	
		AR(1)	ARMA(1,1)	VAR(1)	VAR(2)
Strong	90%	84.0	80.0	82.6	80.9
	70%	82.7	78.7	81.7	80.2
	55%	75.8	73.3	84.3	83.1

Note: The percentage is calculated based on the 300 replications. The result in boldface indicates the maximum value among all the competing models.

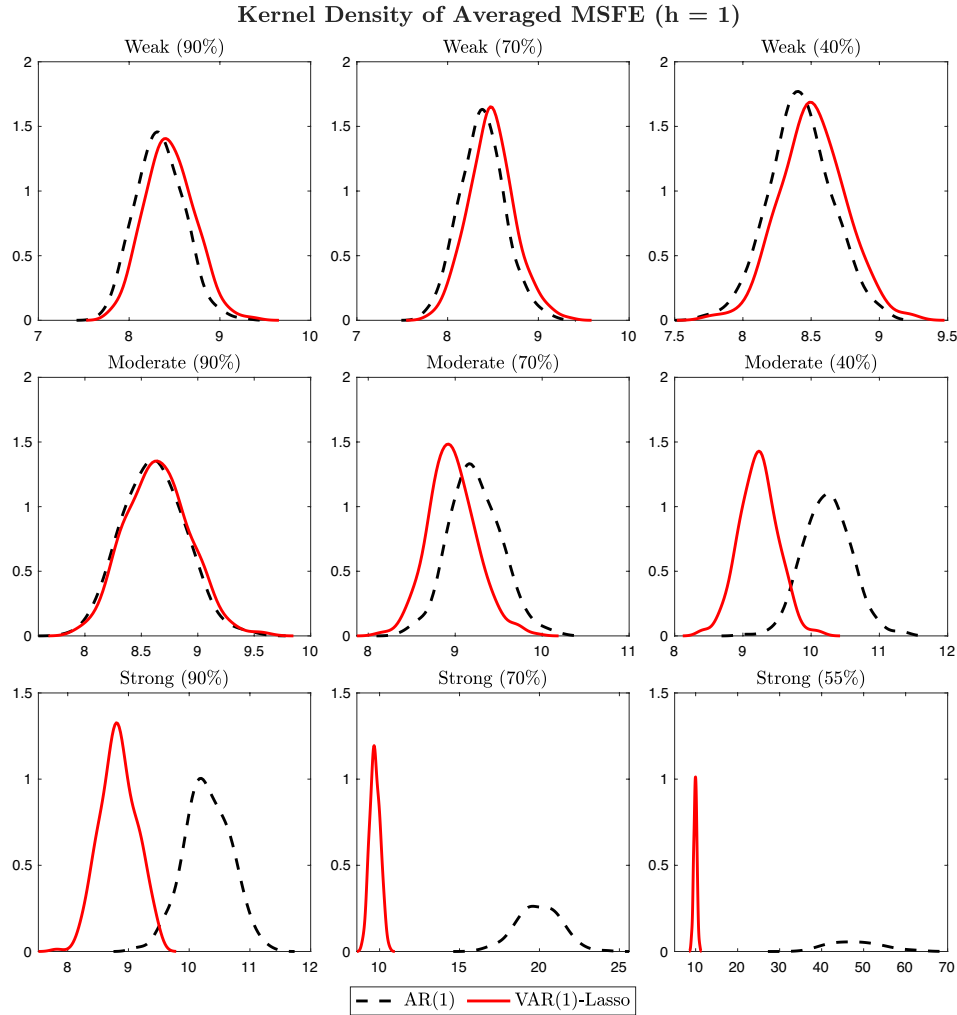


FIGURE 5 | The kernel density plots show distributions of the MSFEs of the AR(1) and VAR(1) models across 300 simulations at a 1-step-ahead horizon under varying connection strengths and sparsity levels. The VAR(1) model is estimated using LASSO.

strength, they are set to the values of the first 30 LGAs in the estimated Sydney VAR system from the entire sample period in all simulations. These values range from -0.275 to 0.263 , with an average value of 0.0154 . The parameter values for the nonzero off-diagonal entries are randomly drawn from a uniform distribution within three interval ranges. For the weak connectivity strength, the interval is $[-0.05, 0.05]$, while for the moderate and strong connectivity strengths, the parameter ranges are $[-0.15, -0.05] \cup [0.05, 0.15]$ and $[-0.35, -0.15] \cup [0.15, 0.35]$, respectively. Finally, the nonzero parameters are randomly distributed across the

matrix Φ_1 . Figure 4 provides a visual illustration of the nine parameter matrices constructed for three levels of connectivity strength and three levels of sparsity.

For each of the nine constructed parameter matrices, we perform 300 simulations resulting in a total of 2700 simulated datasets. Table 6 provides the summary statistics for these datasets. As illustrated by the first two columns in the table, the mean of the simulated series are small and close to zero, while the standard deviations range from about 9 to 23 times the size of the mean values. The magnitude of the standard deviations increases with

the level of connectivity strength. We also observe how the maximum eigenvalue of the coefficient matrix Φ_1 increases with connectivity strength but is inversely related to sparsity. When the connectivity is weak or moderate, the system tends to be stable regardless of the sparsity level. However, when the connectivity strength is high, then low levels of sparsity result in unstable systems with maximum eigenvalues approaching the value of one.

Finally, the last three columns of Table 6 display the average autocorrelation coefficients for the simulated series. We can see that the autocorrelation coefficients at lag one are positively linked to the level of connectivity strength and negatively related to sparsity. Furthermore, at longer time lags, that is, 5 and 10, the average autocorrelations are mostly negligible, except for some cases of high connectivity and low sparsity.

5.2 | Simulation Results

Table 7 presents the frequencies of model inclusion $\bar{f}_m$ based on the out-of-sample results obtained using the SFE loss function. These are computed using the simulation algorithm outlined in Section 2.5. In addition, the results for the absolute forecast error (AFE) loss function can be found in Table A1 in Appendix

A1. As indicated across the top row of the table, the forecasting models evaluated comprise AR(1), ARMA(1, 1), VAR(1), and VAR(2) specifications. The LDVAR models are estimated via LASSO, as described in Section 2.2. Figures 5–8 depict the estimated kernel densities of mean squared forecast errors (MSFEs, across 300 replications) for two superior forecasting models at various forecasting horizons. We also perform the forecasting exercise using the adaptive LASSO method for LDVAR. Our results reveal that the conventional LASSO outperforms the adaptive version in this study. For a more detailed discussion, refer to Appendix A2.

The findings presented in the top panel of Table 7 and Figure 5 demonstrate the importance of connectivity strength for short-term multivariate forecasting. When the connectivity strength is weak, we observe that all VAR-type models underperform the simple AR(1) model, irrespective of the level of sparsity in the lagged parameter matrix. In particular, the MCS results indicate that in over 85% of the simulations with weak connectivity strength, the AR(1) model is found in the set of “superior” forecasting models, whereas the percentage for the VAR(1) model is about 75%. Therefore, weak connectivity, as represented by small off-diagonal elements in the parameter matrix Φ_1 of (23), results in estimation errors that offset the advantages

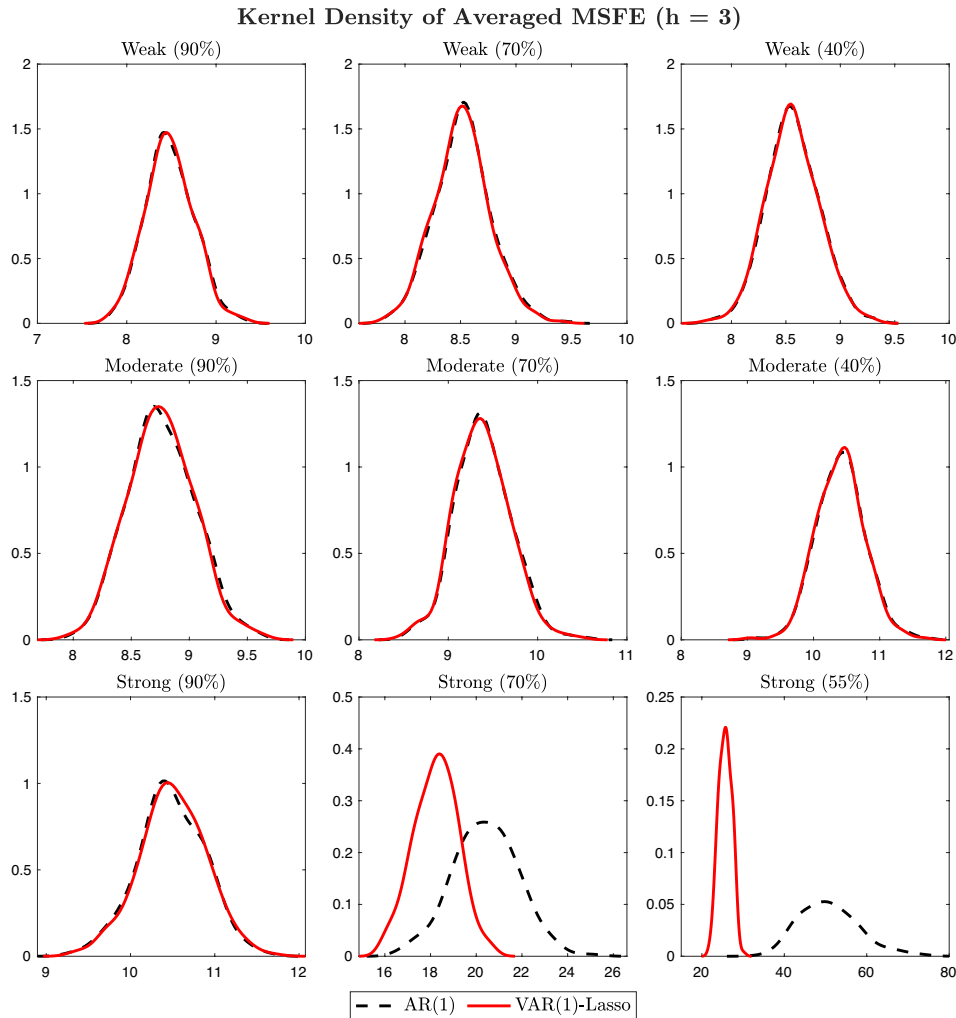


FIGURE 6 | The kernel density plot compares the MSFEs of the AR(1) and VAR(1) models across 300 simulations at a three-step-ahead horizon under varying connection strengths and sparsity levels.

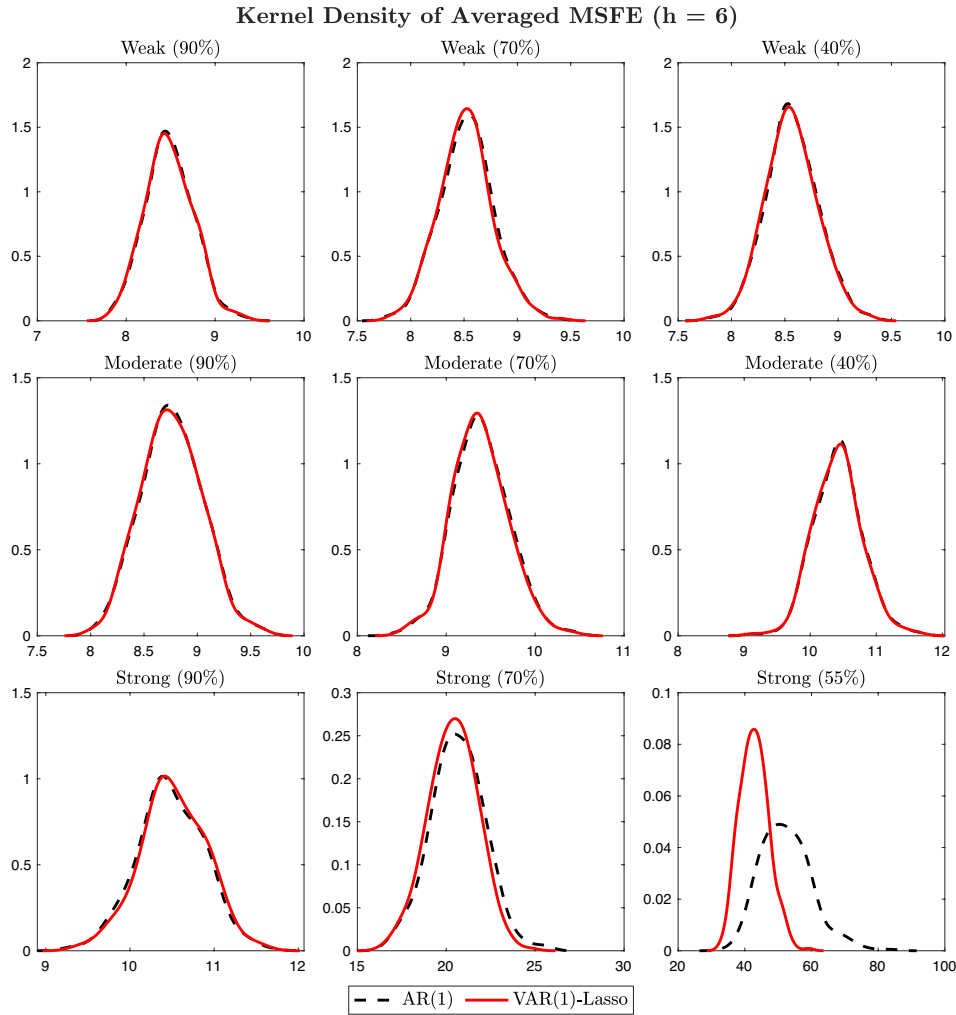


FIGURE 7 | The kernel density plot compares the MSFEs of the AR(1) and VAR(1) models across 300 simulations at a six-step-ahead horizon under varying connection strengths and sparsity levels.

of accounting for interconnectedness between the forecasted variables.

As the connectivity strength increases to moderate and strong levels, the VAR(1) model starts to exhibit forecasting advantage. Furthermore, lower levels of sparsity are associated with stronger performance of the VAR(1) model. For the sparsity levels of 70% and 40% (55%) for moderate (strong) connectivity, the averaged inclusion proportion of the VAR(1) model (i.e., f_m as defined in (2.5)) reaches between 87% and 100%. The percentage for univariate models gradually decreases and eventually reaches 0% under strong connectivity strength and 55% sparsity settings. These patterns are further illustrated in Figure 5. The first row of the figures illustrates the outperformance of AR(1) for weak connectivity, while the second portion of the figure demonstrates moderate and strong levels of connectivity when VAR(1) gains forecasting advantage. The columns of the figure suggest that as sparsity declines, that is, more series are interconnected, the multivariate model produces forecasting gains (except in the case when such connections are small in magnitude).

The forecasting results at midterm prediction horizons exhibit a similar pattern to those of short-term, as presented in the

middle section of Table 7. Although the VAR(1) model with LASSO regression outperforms the AR(1) for $h = 3, 6$ when the sparsity levels are 70% and 40% for moderate (strong) connectivity, the differences between the two models are relatively small for all three connectivity strengths. This is also illustrated in Figures 6 and 7, where the estimated MSFE densities are very similar except in the case of strong connectivity and low sparsity. In fact, when the connectivity is strong reducing the level of sparsity in the parameter matrix, Φ_1 increases the outperformance of the multivariate model relative to the AR(1) specification. On the other hand, when the parameter matrix is sparse enough (i.e., very few linkages exist between the variables), the univariate model provides better predictions.

At longer forecasting horizons, conditional predictions approach their unconditional values. We see that when $h = 12$, the AR(1) model ranks better than VAR(1). This trend is evident from the bottom panel of Table 7, which shows that the AR(1) model consistently outperforms the VAR(1) model in terms of forecasting performance, irrespective of the connectivity strength and connectivity level. An exception to this trend is apparent in the case of strong connectivity strength and 55% sparsity, as depicted in Figure 8. However, it should be noted that the performance of

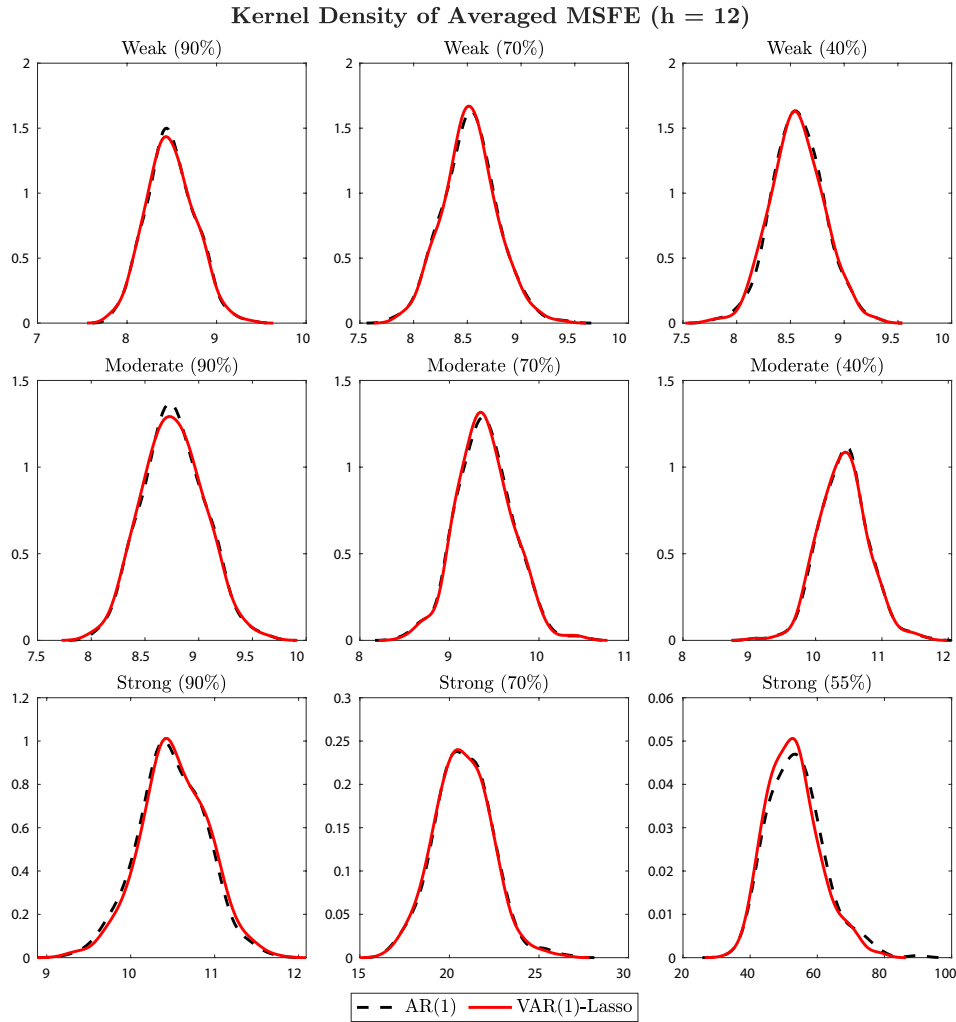


FIGURE 8 | The kernel density plot compares the MSFEs of the AR(1) and VAR(1) models across 300 simulations at a 12-step-ahead horizon under varying connection strengths and sparsity levels.

the VAR(1) model with LASSO is only marginally inferior to that of the AR(1) specification and maintains its stability across all forecast horizons.

In summary, the simulation study provides evidence that both connectivity strength and the level of sparsity have a significant impact on the forecasting performance of the VAR models estimated using LASSO, particularly in short- and midterm forecast horizon scenarios. Nevertheless, the results suggest that connectivity strength appears to exert a greater influence on the forecasting ability than sparsity. These findings can provide valuable insight into forecasting applications where ininterconnectedness between relevant factors needs to be accounted for and especially in large-dimensional systems where both connectivity strength and sparsity need to be considered.

6 | Conclusion

Forecasting house prices is essential for informed decision making in real estate investment, mortgage lending, and policy formulation. Accurate predictions aid homebuyers and sellers in making optimal financial choices, while policymakers rely on

such forecasts to develop housing policies that address affordability and market stability.

In this paper, we investigate the efficacy of incorporating connectivity information to accurately forecast housing price growth rates. We employ an LDVAR model to forecast the growth rate of house prices in a given area, leveraging its historical data and considering all conceivable interactions with other regions within the system. This framework is also extended to incorporate exogenous predictors. The large dimensional model is estimated using shrinkage techniques, and its predictive capability is compared to that of widely recognized benchmark models for predicting house prices, i.e., AR(1) and ARMA(1,1). Given that we model connectedness using LDVAR specifications, we also examine the relationship between the degree of connectedness captured by the models and their predictive accuracy. More specifically, we characterize connectedness by sparsity and connectendess strength. We define sparsity by the ratio of zero elements in the parameter matrices, while connectivity strength is measured by the magnitude of the lagged coefficients. While the statistical characteristics of the LASSO estimators for LDVAR models have been studied, systematic investigations of their out-of-sample forecasting performance and its relationship with the underlying interdependency structure

are limited. To address this gap, we conduct a simulation study that investigates the impact of sparsity and connectivity strength on the forecasting efficacy of LDVAR models.

We consider three housing market systems (within-city areas for Sydney and Melbourne and the city-level markets for China) and conduct rolling window forecasts for each LGA and city within the systems. Our forecasting exercises reveal that the multivariate LDVAR models outperform univariate forecasting specifications when forecasting LGA-level house prices within Sydney and Melbourne. In the case of Chinese city house price returns, univariate AR models consistently outperform the LDVAR specification. Upon closer inspection of the estimated LDVAR parameter matrices, we find that within-city housing markets in Sydney and Melbourne are more densely and strongly connected, whereas the links among Chinese cities are sparse and weak. We conduct a series of robustness analyses to confirm the stability of our empirical findings. These checks include examining the impact of alternative loss functions, considering the inclusion of macroeconomic variables, and computing MCSs using an alternative procedure. The results of these tests support and reinforce our main findings. This raises the question about the role that sparsity and connectivity strength play in forecasting applications of large-dimensional systems and under what configurations of these two parameters can within-system interconnectness lead to forecasting gains. We address this issue by conducting simulation experiments calibrated to reflect the linkages estimated from the data.

The simulation study reveals that the forecasting performance of LDVAR specifications is significantly influenced by both the sparsity of the parameter matrices and the connectivity strength of the system. Interestingly, univariate models, which disregard connectivity information, demonstrate strong performance in cases where interconnectness exists, but the connections are sparse and weak. On the other hand, in the presence of strong connections, incorporating information about interdependencies via LDVAR models becomes important for improving out-of-sample forecasts. In addition, the significance of such relationships is particularly evident when making shorter term forecasts. As the forecast horizon extends to longer periods, the impact of cross-market connectivity gradually diminishes. When the connectivity strength is moderate, connection information proves to be considerably beneficial for short-term forecasts and denser coefficient matrices, but its impact is somewhat diminished for longer forecasting horizons. The effect of sparsity in the LDVAR parameter matrices follows a pattern where the performance of multivariate forecasting models improves, relative to univariate specifications, when the sparsity is low, and vice versa. This effect however is conditional on the connectivity strength, so that when connectivity is weak univariate models outperform at all levels of sparsity considered here. The simulation results are robust to the choice of loss functions (squared and absolute forecast errors) and estimation techniques (LASSO and adaptive LASSO).

Concerning the implications of our study and recommendations, we make the following points. Our results suggest that identifying connectivity within a system does not guarantee improved forecasting performance when detected connections are incorporated into predictive models. Instead, any potential gains in forecasting accuracy depend on the degree of

sparsity and the strength of connectedness within the system. This issue is especially challenging because the sparsity and connectivity strength, which are critical for choosing the right modeling approach, are usually not known before fitting the model. Thus, our recommendation is to acknowledge the potential differences in each housing network and adopt a flexible modeling strategy that involves testing various models and basing decisions on out-of-sample results. Additionally, we recommend that forecasters strive to understand why their chosen model performs best in terms of sparsity and connectivity strength. This insight can guide the ongoing evaluation and adjustments to the model.

Finally, we note some limitations of our study and propose directions for future research. Specifically, we use large-dimensional linear models estimated with shrinkage methods to capture the connectedness within systems of disaggregated house prices. While we account for potential nonlinearity through the use of a rolling window algorithm in forecasting, an alternative approach would be to explicitly incorporate nonlinearity into the model specification, either through a large-dimensional time-varying coefficient network as in Chen et al. (2025) or via a more general nonparametric model estimated using popular machine learning algorithms such as random forests. We leave a systematic comparison between our modeling strategy, and these alternative approaches to future research.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

¹ The LGAs consist of areas greater than a single suburb and represent the lowest tier of government in Australia.

² In Zhu et al. (2017), the authors investigate conditions to ensure the strict stationarity of the model and develop an ordinary least squares-type estimator for the model coefficients.

³ The restricted eigenvalue condition required imposes a lower bound on the minimum eigenvalue of matrix Γ subject to some restrictions on the eigenvectors. It ensures that the predictor variables exhibit sufficient variability and separation, preventing severe multicollinearity and enhancing the stability of the LASSO estimates. When $Np > T$, the sample Gram matrix is singular and the restricted eigenvalue equals zero.

⁴ We have chosen to focus on the VAR(p) model as an example due to its simplicity, as generalizing to more complex models such as VARX(p, k) and ARMA-type models is straightforward.

⁵ An alternative approach to recursive estimation is to use the expanding window estimation. However, this method puts equal weight on all observations from the beginning of the sample to the point of the

forecast, which can lead to biased forecasts when model instability is a concern. Furthermore, in Hansen et al. (2011), rolling window estimation is adopted in the forecasting application to satisfy the stationarity assumptions required by the MCS bootstrap, while recursive or expanding window results are reported only as supplementary pseudo-MCS evidence. Therefore, we employ the rolling window estimation technique, and while expanding window estimation may also be of interest, we leave this approach to future research.

⁶Initially, we additionally examined a 40% sparsity for strong connectivity. However, even with all eigenvalues below one, the resulting data showed traits resembling nonstationary patterns, which is not surprising given findings in Bauwens et al. (2023). Consequently, the minimum sparsity for strong connectivity was adjusted to 55%.

⁷If the LASSO estimator in the initial regression classifies a parameter as zero, then it is not included in the second step of the adaptive LASSO, resulting in a smaller optimization problem. This can lead to computational efficiency and improved accuracy in estimating the nonzero parameters.

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Appendix A

Simulation Study

In this section, we present robustness analyses for the simulation study (1) using the AFE loss function and (2) using the adaptive LASSO estimation method instead of LASSO.

A1 | Simulation Results Based on AFE Loss

Table A1 presents the results of the simulation analysis using the AFE loss function, analogous to Table 7 in Section 5. Notably, the results obtained using the AFE loss function align with the findings derived from the utilization of the SFE loss function. The strength of connectivity has a noteworthy influence on short-term forecasting. In the case when the connectivity strength is weak, the univariate model outperformed the multivariate model. However, as the connectivity strength increases, the multivariate model excels. In terms of long-term forecasting, the univariate AR model generally outperforms the multivariate VAR model, although the difference is typically small. An exception to this pattern occurs for strong connectivity strength combined with 55% sparsity, where the multivariate VAR model performs slightly better. The simulation results present a mixed outcome for midterm forecasting horizons. Specifically, the univariate model demonstrates superior performance when the sparsity level is high, but this advantage is observed primarily when the connectivity strength is weak or moderate. Conversely, the multivariate model exhibits improved performance when the connectivity strength is strong.

A2 | Robustness Check: Adaptive LASSO

The LASSO method has found widespread use in regression analysis for variable selection. Nevertheless, it applies a uniform penalty to all parameters, which might not be the most effective strategy for accurate forecasting. Specifically, applying a more pronounced penalty to parameters near zero and a relatively minor penalty to those of greater magnitude (irrespective of their sign) has the potential to yield a more parsimonious model and enhance forecasting accuracy. This is the precise role fulfilled by the adaptive Lasso technique (Zou 2006). The objective function of the adaptive LASSO is specified as follows:

$$\begin{aligned} \left[\hat{\phi}_0^{(i)}, \hat{\Phi}^{(i)} \right]' = \underset{\phi_0^{(i)}, \Phi^{(i)}}{\operatorname{argmin}} \quad & \frac{1}{2T} \left\| \mathbf{y}^{(i)} - \mathbf{1} \phi_0^{(i)} - \mathbf{Z} \Phi^{(i)} \right\|_{\ell_2}^2 \\ & + \lambda \sum_{i,j} \omega_{ij} |\phi_{ij}^{(i)}|, \end{aligned} \quad (\text{C1})$$

where $\omega_{ij} = |\tilde{\phi}_{ij}^{(i)}|^{-\gamma}$ is the weight factor for coefficient $\phi_{ij}^{(i)}$, with $i = 1, \dots, N$, $j = 1, \dots, p$. The parameter γ is a positive constant, taking typical values of 0.5, 1, and 2. The value $\tilde{\phi}_{ij}^{(i)}$ is an estimate of $\phi_{ij}^{(i)}$ obtained from an initial regression method, such as OLS, Ridge, or LASSO.⁷ This weight function establishes an inverse relationship with the absolute magnitude of the coefficient. Consequently, the adaptive LASSO technique facilitates more assertive reduction of coefficients associated with less significant variables.

Figures A.1–A.4 depict the average MSFEs for the VAR(1) model using LASSO and adaptive Lasso across various forecast horizons. The plot reveals that LASSO exhibits comparable forecasting performance to adaptive Lasso, with instances where LASSO outperforms its counterpart.

TABLE A1 | The averaged percentage of one model being included in a 90% MCS at different forecasting horizons, built on the AFE loss function.

		Univariate		Multivariate	
Strength	Sparsity	AR(1)	ARMA(1,1)	VAR(1)	VAR(2)
$h = 1$					
Weak	90%	85.8	78.1	77.3	76.0
	70%	86.3	78.3	77.3	75.9
	40%	86.2	78.3	78.5	75.8
Moderate	90%	83.4	75.3	80.3	76.0
	70%	77.9	70.5	86.2	75.8
	40%	64.0	58.9	92.5	73.7
Strong	90%	42.8	39.6	92.7	77.3
	70%	1.9	1.9	97.1	65.6
	55%	0.0	0.0	99.2	44.6
$h = 3$					
Weak	90%	83.1	78.0	82.2	82.3
	70%	81.3	77.9	83.3	81.6
	40%	83.1	78.1	83.3	82.6
Moderate	90%	83.0	78.4	83.0	82.5
	70%	81.9	77.9	83.3	81.7
	40%	82.5	78.2	83.9	81.8
Strong	90%	82.6	78.4	83.6	81.6
	70%	60.8	57.7	90.8	78.1
	55%	4.4	4.4	97.5	61.6
$h = 6$					
Weak	90%	82.6	79.3	82.5	81.6
	70%	82.3	78.9	82.7	81.6
	40%	83.2	78.7	83.5	81.7
Moderate	90%	82.6	79.1	82.5	81.5
	70%	82.5	79.4	83.1	81.6
	40%	82.7	78.4	83.3	81.8
Strong	90%	82.2	78.6	82.3	81.8
	70%	81.4	77.8	83.3	81.3
	55%	42.1	41.0	94.4	75.7
$h = 12$					
Weak	90%	83.2	79.9	82.4	80.7
	70%	82.9	79.4	81.6	80.7
	40%	82.9	79.9	82.0	80.7
Moderate	90%	82.7	79.7	82.2	81.2
	70%	83.6	79.8	82.6	81.3
	40%	82.7	79.7	82.6	81.6
Strong	90%	83.7	80.1	82.2	81.0
	70%	83.2	79.5	81.8	80.7
	55%	76.8	73.9	83.8	83.0

Note: The percentage is calculated based on the 300 replications. The result in boldface indicates the maximum value among all the competing models.

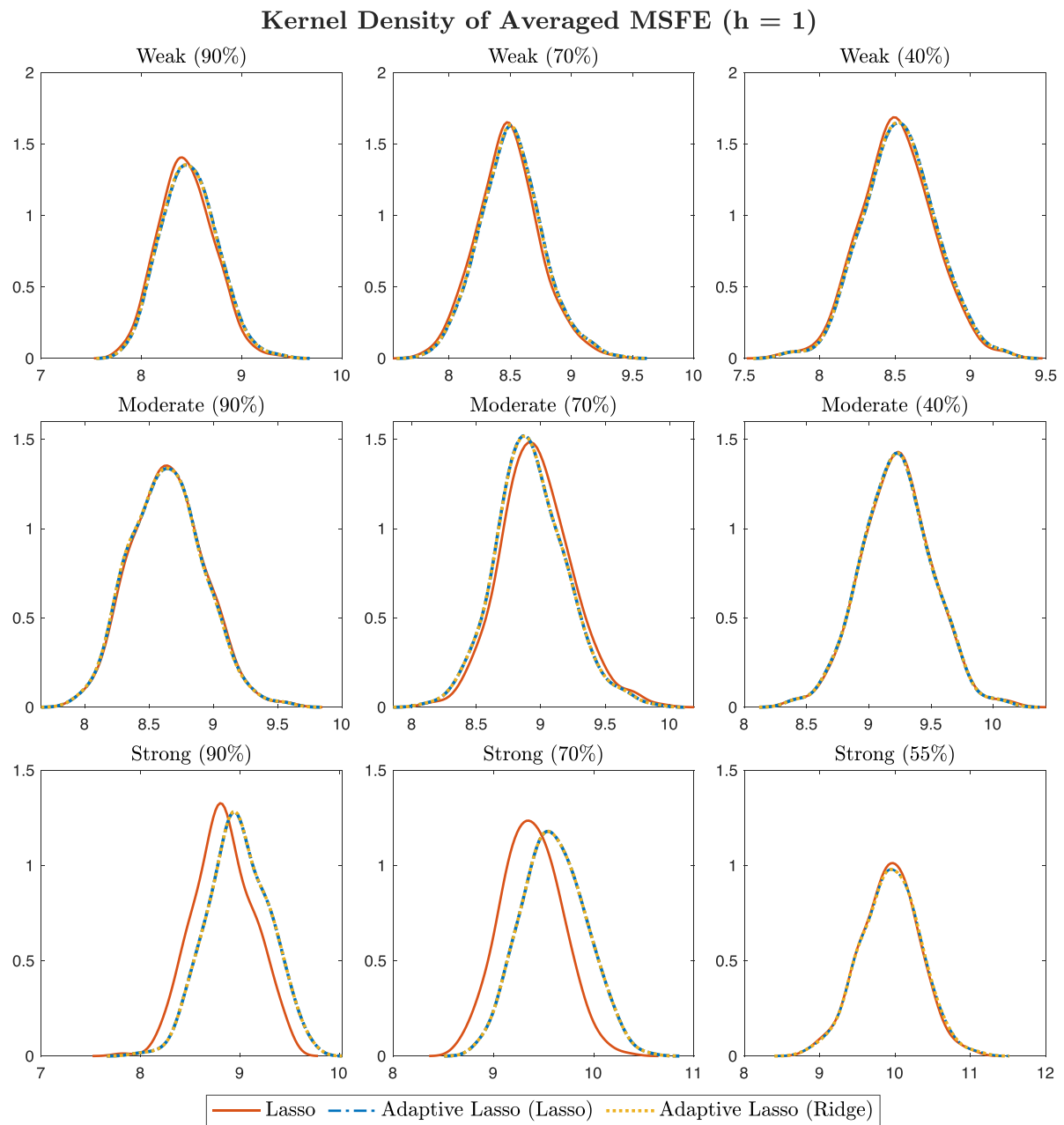


FIGURE A.1 | The kernel density plots illustrate the average MSFEs for one-step-ahead forecasts of the VAR(1) model with different estimation methods (LASSO and adaptive LASSO). The parentheses indicate the initial regression method employed in the adaptive Lasso approach.

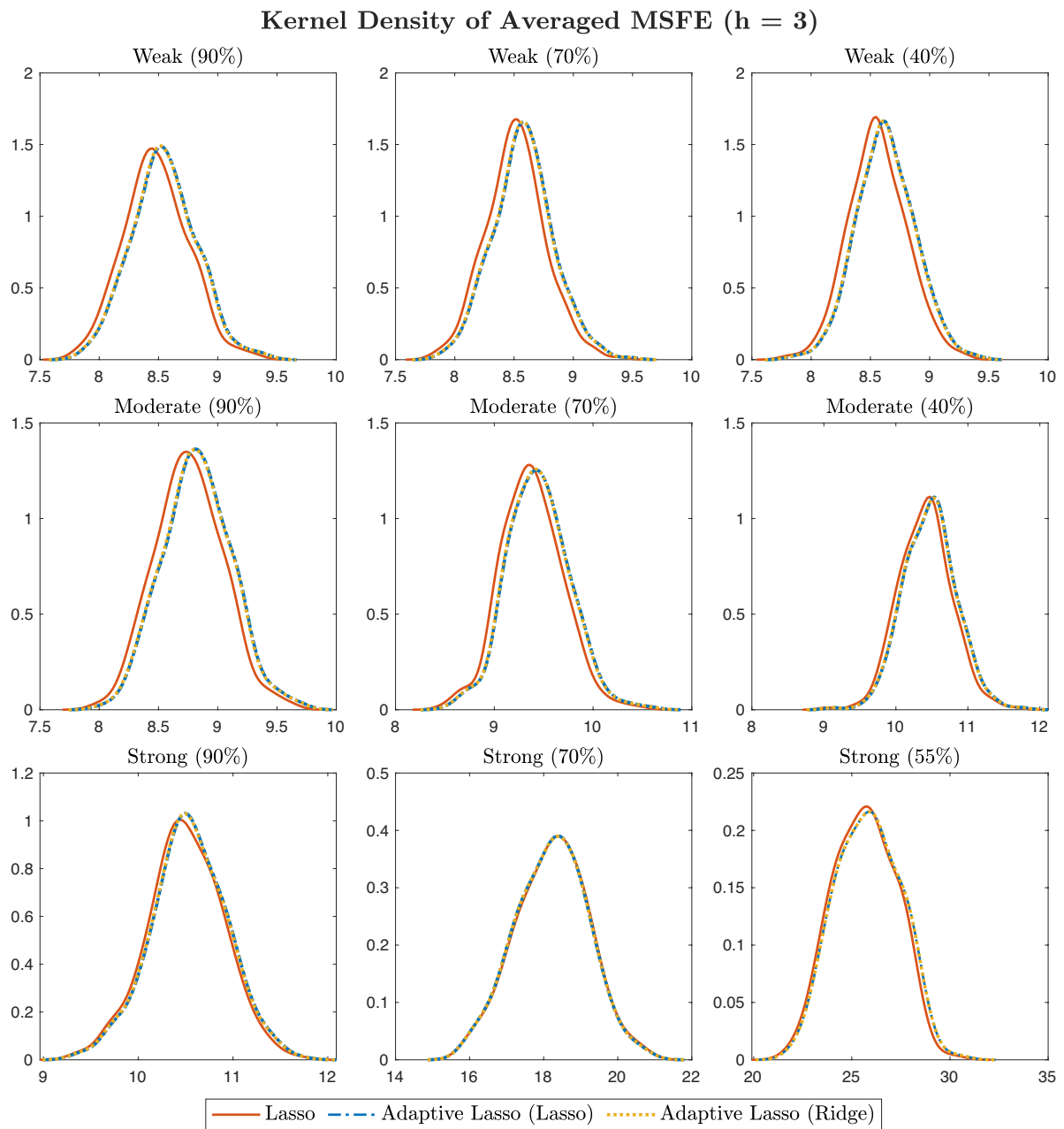


FIGURE A.2 | The kernel density plots illustrate the average MSFEs for three-step-ahead forecasts of the VAR(1) model with different estimation methods (LASSO and adaptive LASSO). The parentheses indicate the initial regression method employed in the adaptive Lasso approach.

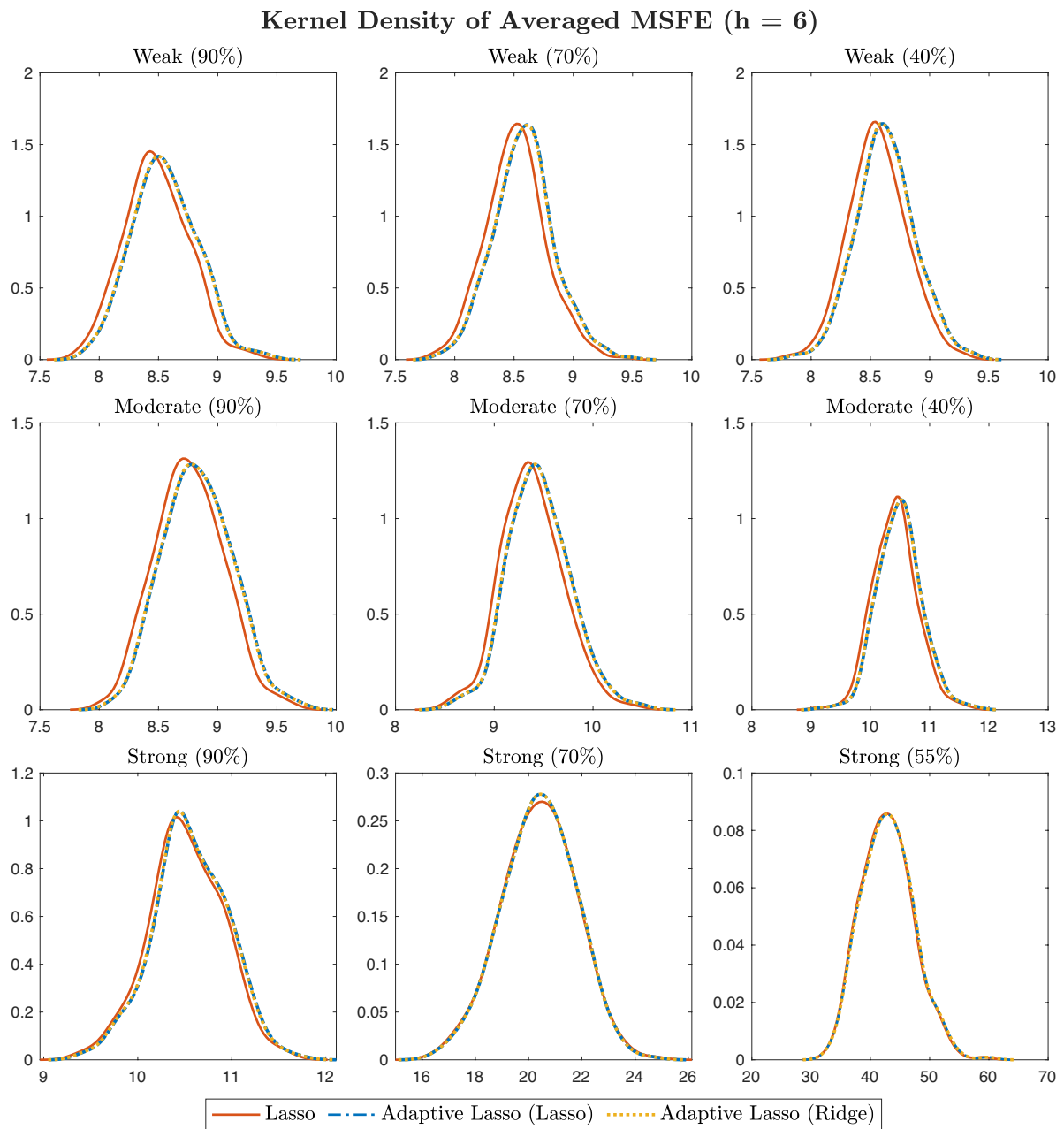


FIGURE A.3 | The kernel density plots illustrate the average MSFEs for six-step-ahead forecasts of the VAR(1) model with different estimation methods (LASSO and adaptive LASSO). The parentheses indicate the initial regression method employed in the adaptive Lasso approach.

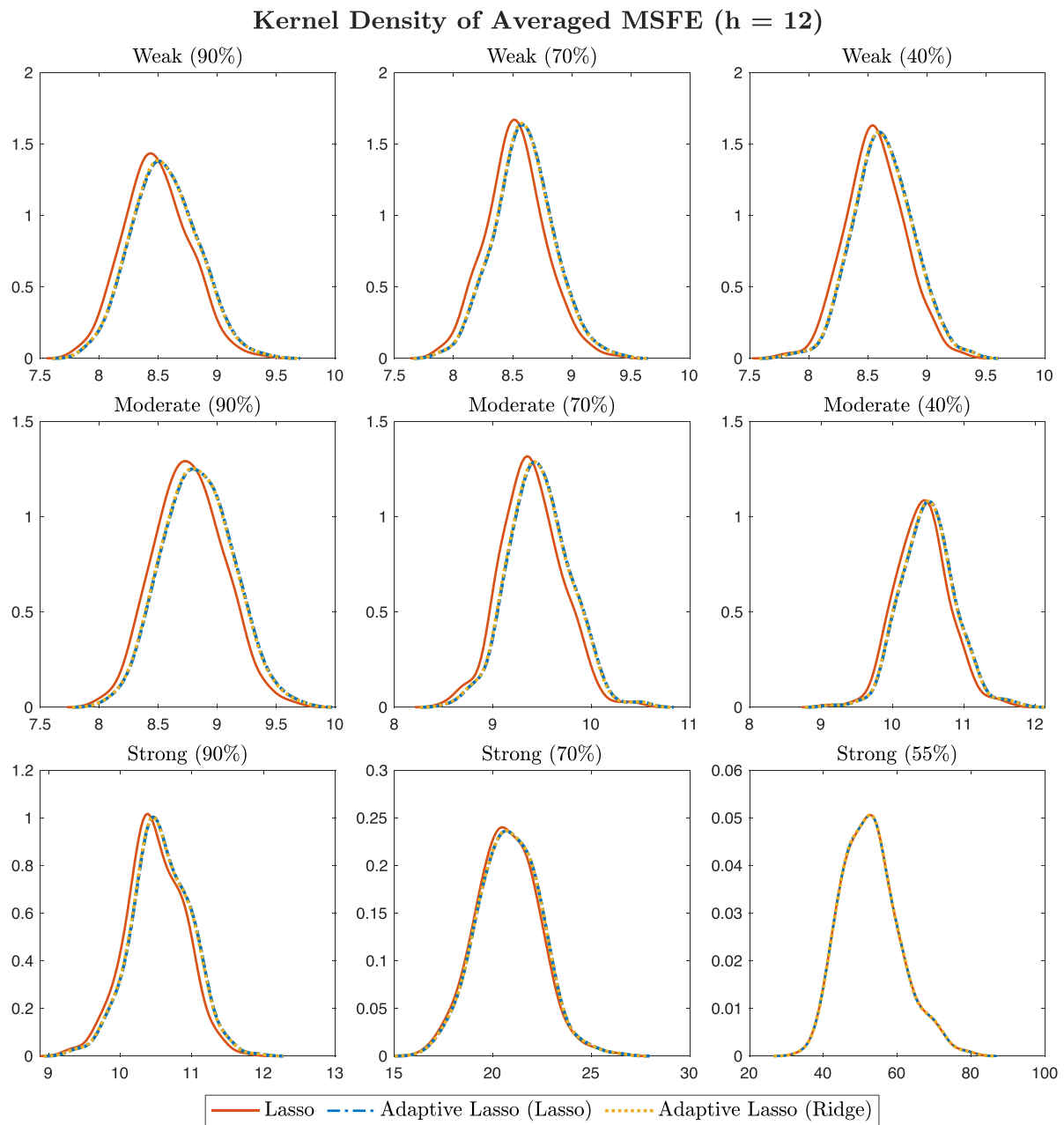


FIGURE A.4 | The kernel density plots illustrate the average MSFEs for six-step-ahead forecasts of the VAR(1) model with different estimation methods (LASSO and adaptive LASSO). The parentheses indicate the initial regression method employed in the adaptive Lasso approach.

Appendix B

Lists of LGAs and Cities

TABLE B1 | List of LGAs included in the empirical analysis for Sydney and Melbourne; LGAs are grouped by region.

Sydney		Melbourne	
Western City	Blue Mountains	Eastern Region	Knox
	Hawkesbury		Manningham
	Penrith		Maroondah
	Camden		Monash
	Campbelltown		Whitehorse
	Fairfield		Yarra Ranges
	Liverpool	Inner Metro Region	Melbourne
Central City	Wollondilly		Port Phillip
	Blacktown		Yarra
	Cumberland	Inner South East Region	Bayside
	Parramatta		Boroondara
Eastern City	The Hills Shire		Glen Eira
	Burwood		Stonnington
	Canada Bay	Northern Region	Banyule
	Inner West		Darebin
	Randwick		Hume
	Strathfield		Moreland
	Woollahra	Southern Region	Nillumbik
North District	Waverley		Whittlesea
	Sydney		Cardinia
	Hornsby		Casey
	Hunters Hill		Frankston
	Kuringgai		Greater Dandenong
	Lane Cove		Kingston
South District	Northern Beaches		Mornington Peninsula
	Mosman	Western Region	Brimbank
	Willoughby		Hobsons Bay
	Ryde		Maribyrnong
	North Sydney		Melton
	Georges River		Moonee Valley
	Canterbury–Bankstown		Wyndham
	Sutherland		

TABLE B2 | List of Tiers 1–3 cities for the Chinese housing market.

Tier 1 cities			
Beijing	Guangzhou	Shanghai	Shenzhen
Tier 2 cities			
Changchun	Changsha	Chengdu	Chongqing
Dalian	Haikou	Hangzhou	Harbin
Hefei	Hohhot	Jinan	Nanchang
Nanjing	Nanning	Ningbo	Qingdao
Shenyang	Shijiazhuang	Suzhou	Tianjin
Wenzhou	Wuxi	Xi'An	Xiamen
Xining	Zhengzhou		
Tier 3 cities			
Anqing	Anshan	Baoding	Baotou
Bengbu	Changde	Changzhou	Chuzhou
Dandong	Deyang	Dongguan	HuaiAn
Huzhou	Jiaxing	Jieyang	Jingdezhen
Jinhua	Jiujiang	Kaifeng	Langfang
Leshan	Lianyungang	Luohe	Luoyang
Luzhou	Mianyang	Nanchong	Nantong
Nanyang	Ningde	Qinhuangdao	Quanzhou
Rizhao	Shangrao	Shantou	Shaoxing
Songyuan	Suqian	Taizhou	Tangshan
Tieling	Wuhu	Huludao	Xingtai
Xinxiang	Xuancheng	Xuzhou	Yancheng
Yangzhou	Yichun	Yingkou	Zaozhuang
Zhangjiakou	Zhangzhou	Zhaoqing	Zhenjiang
Zhongshan	Zhumadian		